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★ INTEGRAL CALCULUS ★

★ Gamma function

Gamma function Γn is defined by the improper integral

$$\int_0^\infty e^{-x} \cdot x^{n-1}\, dx \quad , \quad \text{where } n>0$$

Hence

$$\boxed{\Gamma n = \int_0^\infty e^{-x}\, x^{n-1}\, dx}$$

Note

$$\int_0^\infty e^{-kx} \cdot x^{n-1}\, dx = \frac{\Gamma n}{k^n}$$

It is also known as Euler's integral of second kind.

★ Properties of Gamma function

① $\Gamma n = (n-1)!$ for $n>0$ OR $\Gamma(n+1) = n!$ $n \geq 0$ or $n+1 \geq 1$

② $\Gamma(n+1) = n\,\Gamma n$ } for $n>0$, Real no.

③ $\Gamma n = \frac{\Gamma(n+1)}{n}$, for n is -ve fraction

④ $\Gamma n \cdot \Gamma(1-n) = \frac{\pi}{\sin n\pi}$

⑤ $\Gamma 1 = 1$

$\Gamma 2 = 1$

⑥ $\Gamma\frac{1}{2} = \sqrt{\pi}$

⑦ $\Gamma n = \infty$ if n is -ve integer or n is zero

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Ex: Show that $\overline{n+1} = n\,\overline{n}$ for $n > 0$

Soln,

By defⁿ of Gamma function

$$\overline{n+1} = \int_0^\infty e^{-x} \cdot x^{(n+1)-1}\, dx$$

$$= \int_0^\infty e^{-x}\, x^n\, dx$$

II I

Integrating By parts we get

$$= \left\{ x^n \int e^{-x}\, dx - \int \left[\frac{d}{dx}(x^n) \cdot \int e^{-x}\, dx \right] dx \right\}_0^\infty$$

$$= \left[x^n (-e^{-x}) \right]_0^\infty - \int_0^\infty \left[n x^{n-1} (-e^{-x}) \right] dx$$

$$= [0 + 0] + n \int_0^\infty e^{-x} \cdot x^{n-1}\, dx$$

$$= n \cdot \overline{n}$$

Hence

$$\boxed{\overline{n+1} = n\,\overline{n}}$$

H.P.

$\Rightarrow$

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Ex: Show that $\lceil n+1 = n!$, for $n>0$

Soln: For $n>0$, we have

$$\lceil n+1 = n \lceil n$$
$$= n \lceil (n-1)+1$$
$$= n (n-1) \lceil n-1$$
$$= n (n-1) (n-2) \lceil n-2$$
$$= n (n-1) (n-2) (n-3) \lceil n-3$$
$$= n (n-1) (n-2) (n-3) \cdots 3 \cdot 2 \cdot 1$$
$$= n!$$

Hence $\boxed{\lceil n+1 = n!}$

Ex: Show that $\lceil 1 = 1$, & $\lceil 2 = 1$

Soln we know that

$$\lceil n = (n-1)!$$

$\therefore$ put $n=1 \Rightarrow \lceil 1 = (1-1)! = 0! = 1$

put $n=2 \Rightarrow \lceil 2 = (2-1)! = 1! = 1$

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Ex: prove that $\overline{|n} = \infty$ if n is -ve or zero

Soln. we know that

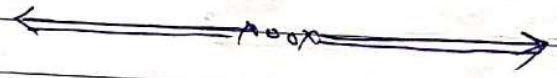
$$\overline{|n+1} = \frac{\overline{|n+1}}{n} \quad \text{---(1)}$$

put $n=0$ $\Rightarrow$ $\overline{|0} = \frac{\overline{|0+1}}{0} = \frac{1}{0} = \infty$

put $n=-1$ $\Rightarrow$ $\overline{|-1} = \frac{\overline{|-1+1}}{-1} = \frac{\overline{|0}}{-1} = \frac{\infty}{-1} = \infty$

put $n=-2$ $\Rightarrow$ $\overline{|-2} = \frac{\overline{|-2+1}}{-2} = \frac{\overline{|-1}}{-2} = \frac{\infty}{-2} = \infty$

and so on.



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Ex : Find the values of

(i) $\Gamma(-5/2)$,

(2) $\Gamma(7)$, (3) $\Gamma(8/3)$

Soln : (1) We know that

$$\Gamma(n+1) = n\,\Gamma(n)$$

$$\Rightarrow \Gamma(n) = \frac{\Gamma(n+1)}{n}$$

Put $n = -5/2$

$$\therefore \Gamma(-5/2) = \frac{\Gamma(-5/2+1)}{-5/2} = \frac{-2}{5}\,\Gamma(-3/2)$$

$$= \frac{-2}{5}\,\frac{\Gamma(-3/2+1)}{(-3/2)}$$

$$= \frac{4}{15}\,\Gamma(-1/2)$$

$$= \frac{4}{15}\,\frac{\Gamma(-1/2+1)}{(-1/2)}$$

$$= -\frac{8}{15}\,\Gamma(1/2)$$

$$= -\frac{8}{15}\sqrt{\pi}$$

Hence

$$\Gamma(-5/2) = -\frac{8}{15}\sqrt{\pi}$$

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Soln ② We know that $\lfloor n+1 = n!$ if n is positive integer

$\therefore$ put $n=6$

$$\lfloor 7 = \lfloor 6+1$$
$$= 6!$$
$$= 6\times5\times4\times3\times2\times1$$
$$= 720$$

Soln ③ We know that

$$\lfloor n+1 = n \lfloor n \quad \text{if } n \text{ is positive fraction}$$

$$\therefore \lfloor 8/3 = \lfloor \frac{5}{3}+1$$
$$= \frac{5}{3} \lfloor 5/3$$
$$= \frac{5}{3} \lfloor \frac{2}{3}+1$$
$$= \frac{5}{3}\cdot\frac{2}{3}\cdot\lfloor 2/3 = \frac{10}{9}\lfloor \frac{2}{3}$$

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Ex : Evaluate

(i) $\int_0^\infty e^{-4x} \cdot x^{3/2}\, dx$

(ii) $\int_0^\infty e^{-x^2} \cdot x^2\, dx$

(Show $\int_0^\infty e^{-kx} \cdot x^{n-1}\, dx = \frac{\Gamma n}{k^n}$. Hence evaluate.)

Soln (i)

let

$$I = \int_0^\infty e^{-4x} \cdot x^{3/2}\, dx$$

put $4x = t$

$\Rightarrow dx = \frac{dt}{4}$ and as $x=0 \Rightarrow t=0$, $x=\infty \Rightarrow t=\infty$

$$\therefore I = \int_{t=0}^\infty e^{-t} \cdot \left(\frac{t}{4}\right)^{3/2} \cdot \frac{dt}{4}$$

$$= \int_0^\infty e^{-t} \cdot t^{3/2} \cdot \frac{dt}{(4)^{5/2}}$$

$$= \frac{1}{(4)^{5/2}} \cdot \int_0^\infty e^{-t} \cdot t^{3/2}\, dt$$

$$= \frac{1}{2^5} \cdot \int_0^\infty e^{-t} \cdot t^{5/2-1}\, dt$$

$$= \frac{1}{32} \overline{|5/2} = \frac{1}{32} \cdot \frac{3}{2} \overline{|3/2}$$

$$= \frac{1}{32} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{3\sqrt{\pi}}{128}$$

$\overline{|1} = 1$, $\overline{|n+1} = n!$, $\overline{|n} = (n-1)!$

$\overline{|3/2} = \frac{1}{2}\sqrt{\pi}$

by defn $\Gamma n = \int_0^\infty e^{-t} t^{n-1}\, dt$

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$$= \frac{5}{32} \cdot \Gamma(3/2+1)$$

$$= \frac{1}{32} \cdot \frac{3}{2} \Gamma(3/2) \quad \text{---} \because \Gamma(n+1) = n\Gamma(n)$$

$$= \frac{3}{2^5 32} \Gamma\left(\frac{1}{2}+1\right)$$

$$= \frac{3}{2 \times 32 \times 2} \cdot \Gamma\frac{1}{2}$$

$$= \frac{3}{128} \sqrt{\pi}$$

$$\therefore \boxed{\int_0^\infty e^{-4x} \cdot x^{3/2}\, dx = \frac{3\sqrt{\pi}}{128}}$$

Soln (2) Let $I = \int_0^\infty e^{-x^2} \cdot x^2\, dx$

put $x^2 = t$

$\Rightarrow 2x\, dx = dt$ i.e. $dx = \frac{dt}{2x} = \frac{dt}{2t^{1/2}}$

and as $x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$$\therefore I = \int_0^\infty e^{-t} \cdot t \cdot \frac{dt}{2t^{1/2}}$$

$$= \frac{1}{2} \int_0^\infty e^{-t} \cdot t \cdot t^{-1/2} \cdot dt$$

$$= \frac{1}{2} \int_0^\infty e^{-t} \cdot t^{1/2} \cdot dt$$

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$$= \frac{1}{2} \int_0^\infty e^{-t} \cdot t^{\frac{3}{2}-1} \, dt$$

$$= \frac{1}{2} \overline{|3/2}$$

$$= \frac{1}{2} \overline{|1/2+1}$$

$$= \frac{1}{2} \cdot \frac{1}{2} \overline{|1/2} \qquad \because \overline{|n+1} = n \overline{|n}$$

$$= \frac{\sqrt{\pi}}{4}$$

$$\therefore \boxed{\int_0^\infty e^{-x^2} \cdot x^2 \, dx = \frac{\sqrt{\pi}}{4}}$$

Ex: Show that $\int_0^\infty e^{-kx} \cdot x^{n-1} \, dx = \frac{\overline{|n}}{k^n}$,

Soln: Let $I = \int_0^\infty e^{-kx} x^{n-1} \, dx$

put

$$kx = t \quad \Rightarrow dx = \frac{dt}{k}$$

as $x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$$\therefore I = \int_0^\infty e^{-t} \cdot \left(\frac{t}{k}\right)^{n-1} \cdot \frac{dt}{k}$$

$$= \frac{1}{k^n} \int_0^\infty e^{-t} t^{n-1} \, dt$$

$$= \frac{\overline{|n}}{k^n}$$ — M.P.

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Ex: Show that

$$\int_0^\infty \frac{x^c}{c^x}\,dx = \frac{\overline{|c+1}}{(\log c)^{c+1}}, \quad \text{where } c>1$$

Soln:

let $I = \int_0^\infty \frac{x^c}{c^x}\,dx$

$= \int_0^\infty x^c \cdot \bar{c}^x\,dx$

put $c^{-x} = e^{-t}$ $\Rightarrow x \log c = t \log_e e = t$

$\Rightarrow x = \frac{t}{\log c}$ $\Rightarrow dx = \frac{dt}{\log c}$

as $x=0 \Rightarrow t=0$

as $x=\infty \Rightarrow t=\infty$

$x=\infty,\ c^0 = e^t \Rightarrow e^t = 1 \Rightarrow t=0$

$x=\infty\ \ c^\infty = e^t \Rightarrow \infty = e^t \Rightarrow t=\infty$

$\therefore I = \int_0^\infty \bar{e}^t \cdot \left(\frac{t}{\log c}\right)^c \cdot \frac{dt}{\log c}$

$= \int_0^\infty \bar{e}^t \cdot t^c \cdot \frac{dt}{(\log c)^{c+1}}$

$= \frac{1}{(\log c)^{c+1}} \int_0^\infty \bar{e}^t \cdot t^{(c+1)-1}\,dt$

$= \frac{\overline{|c+1}}{(\log c)^{c+1}}$

$\therefore \int_0^\infty \frac{x^c}{c^x}\,dx = \frac{\overline{|c+1}}{(\log c)^{c+1}}$ Hence

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Ex: Evaluate $\int_0^\infty \sqrt{x} \cdot e^{-x^2}\,dx \cdot \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}\,dx$

Soln: Let

$$I = \int_0^\infty \sqrt{x}\, e^{-x^2}\,dx \times \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}\,dx \quad \text{---(1)}$$

Put $x^2 = t \Rightarrow x = t^{1/2}$

$$\therefore dx = \frac{1}{2} t^{-1/2} \cdot dt$$

as $x=0 \Rightarrow t=0$

$x=\infty \Rightarrow t=\infty$

$$\therefore I = \left[\int_0^\infty \sqrt{t^{1/2}} \cdot e^{-t} \cdot \frac{1}{2} t^{-1/2} \cdot dt\right] \times \left[\int_0^\infty \frac{e^{-t}}{\sqrt{t^{1/2}}} \cdot \frac{1}{2} t^{-1/2} \cdot dt\right]$$

$$= \frac{1}{2}\left[\int_0^\infty e^{-t} \cdot t^{1/4} \cdot t^{-1/2}\,dt\right] \times \left[\frac{1}{2}\int_0^\infty e^{-t}\, t^{-1/4}\, t^{-1/2} \cdot dt\right]$$

$$= \frac{1}{4}\left[\int_0^\infty e^{-t}\, t^{-1/4}\,dt\right] \cdot \left[\int_0^\infty e^{-t} \cdot t^{-3/4}\,dt\right]$$

$$= \frac{1}{4}\left(\int_0^\infty e^{-t} \cdot t^{3/4-1}\,dt\right)\left(\int_0^\infty e^{-t}\, t^{1/4-1}\,dt\right)$$

$$= \frac{1}{4} \lceil 3/4 \cdot \lceil 1/4$$

$$= \frac{1}{4} \lceil 3/4 * \lceil 1-3/4$$

$$= \frac{1}{4} \frac{\pi}{\sin \frac{3\pi}{4}} = \frac{1}{4} \frac{\pi}{(1/\sqrt{2})} = \frac{\pi}{2\sqrt{2}}$$

$\because \lceil n \lceil 1-n = \frac{\pi}{\sin n\pi}$

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Ex: Show that

$$\overline{n} = \int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx, \quad n>0$$

Soln: By defn of Gamma function

$$\overline{n} = \int_{t=0}^{\infty} e^{-t} \cdot t^{n-1}\, dt \qquad \text{---(1)}$$

Put $t = \log(1/x) \Rightarrow \frac{1}{x} = e^t \Rightarrow x = e^{-t}$

$\Rightarrow -e^{-t}\, dt = dx$

$\Rightarrow dt = \frac{dx}{-e^{-t}} = -dx$

as $t=0 \Rightarrow x=1$

$t=\infty \Rightarrow x=0$

(1) $\Rightarrow \therefore \overline{n} = \int_{x=1}^{x=0} \left(\log \frac{1}{x}\right)^{n-1} \cdot (-dx)$

$$= \int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx$$

$$\therefore \boxed{\overline{n} = \int_0^1 \left(\log \frac{1}{x}\right)^{n-1} dx}$$

OR

Let $I = \int_0^1 (\log 1/x)^{n-1}\, dx$ ---(1)

put $\log 1/x = t$

$\Rightarrow x = e^{-t}$

$\therefore dx = -e^{-t}\, dt$

as $x=0 \Rightarrow t=\infty$

$x=1 \Rightarrow t=0$

$\therefore I = \int_{\infty}^{0} t^{n-1} \cdot (-e^{-t})\, dt$

$= \int_0^{\infty} e^{-t} \cdot t^{n-1}\, dt$

Hyp. $= \overline{n}$

Thus

$$\overline{n} = \int_0^1 (\log 1/x)^{n-1}\, dx$$

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Ex: Evaluate: $\int_0^1 x^3 \log\left(\frac{1}{x}\right)^4 dx$

Soln: let $I = \int_0^1 x^3 \log\left(\frac{1}{x}\right)^4 dx$

$$= 4 \int_0^1 x^3 \log\left(\frac{1}{x}\right) \cdot dx \quad \text{---} (1)$$

put $t = \log\left(\frac{1}{x}\right) \Rightarrow x \cdot x = e^{-t}$

$\therefore dx = -e^{-t} \cdot dt$

as $x=0 \Rightarrow t=\infty$ and as $x=1 \Rightarrow t=0$

$$\therefore I = 4 \int_\infty^0 e^{-3t} \cdot t \cdot (-e^{-t}) dt$$

$$= 4 \int_0^\infty e^{-4t} \cdot t \, dt \qquad \because \int_a^b = -\int_b^a$$

$$= 4 \int_0^\infty e^{-4t} \cdot t^{2-1} \, dt$$

$$= 4 \left(\frac{\lfloor 2}{4^2}\right) \qquad \because \int_0^\infty e^{-kt} \cdot t^{n-1} dt = \frac{\lfloor n}{k^n}$$

$$= \frac{1}{4}$$

$$\therefore \int_0^1 x^3 \log\left(\frac{1}{x}\right)^4 \cdot dx = \frac{1}{4}.$$

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✓ Ex: with certain limitations on the value of a, b, m and n, prove that

$$\int_0^\infty \int_0^\infty e^{-(ax^2+by^2)} x^{2m-1} \cdot y^{2n-1}\, dx\, dy = \frac{\lceil m \lceil n}{4 a^m \cdot b^n}$$

Soln :⇒

let $$I = \int_0^\infty \int_0^\infty e^{-(ax^2+by^2)} \cdot x^{2m-1} \cdot y^{2n-1}\, dx\, dy$$

$$= \left(\int_0^\infty e^{-ax^2} \cdot x^{2m-1}\, dx\right) \cdot \left(\int_0^\infty e^{-by^2} y^{2n-1}\, dy\right)$$

$$= I_1 \times I_2 \quad \text{——(1)}$$

Here $$I_1 = \int_0^\infty e^{-ax^2} x^{2m-1}\, dx$$

Put $ax^2 = t \Rightarrow x = (t/a)^{1/2}$

$$dx = \frac{1}{2a}\left(\frac{t}{a}\right)^{-1/2} dt$$

$$= \frac{t^{-1/2}}{2\, a^{1/2}} \cdot dt$$

as $x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$$I_1 = \int_0^\infty e^{-t} \cdot \left(\frac{t}{a}\right)^{\frac{2m-1}{2}} \cdot \frac{t^{-1/2}}{2\, a^{1/2}}\, dt$$

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$$= \frac{1}{2\, a^{1/2} \cdot a^{\frac{2m-1}{2}}} \int_0^\infty e^{-t} \cdot t^{\frac{(2m-1)}{2}} \cdot t^{-1/2}\, dt$$

$$= \frac{1}{2\, a^{m}} \int_0^\infty e^{-t} \cdot t^{(m-1/2-1/2)}\, dt$$

$$= \frac{1}{2a^m} \int_0^\infty e^{-t}\, t^{m-1}\, dt$$

$$I_1 = \frac{\lfloor m}{2a^m}$$

Similarly $I_2 = \int_0^\infty e^{-by^2} \cdot y^{2n-1}\, dy = \frac{\lfloor n}{2b^n}$.

Hence

$$I = I_1 * I_2$$

$$= \frac{1}{4} \frac{\lfloor m \cdot \lfloor n}{a^m \cdot b^n}$$

H.P.

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H.W :

① Find the values of (i) $\overline{-1/2}$ Ans $-2\sqrt{\pi}$

(ii) $\overline{-3/2}$ Ans $\frac{4}{3}\sqrt{\pi}$

(iii) $\overline{-9/2}$ Ans $\frac{-32}{315}\sqrt{\pi}$

Hint $\overline{n} = \frac{\overline{n+1}}{n}$

② prove that $\int_0^1 x^{n-1}\left(\log\frac{1}{x}\right)^{m-1} dx = \frac{\overline{m}}{n^m}$

③ Evaluate $\int_0^\infty x^6 e^{-2x} dx$ Ans $\frac{45}{8}$

Hint : put $2x = t$

④ Evaluate $\int_0^\infty x^n e^{-\sqrt{ax}} dx$ Ans $\frac{2\overline{2n+2}}{a^{n+1}}$

Hint put $\sqrt{ax} = t$

⑤ Evaluate : $\int_0^\infty 3^{-4x^2} dx$ Ans $\frac{\sqrt{\pi}}{4\sqrt{\log 3}}$

Hint put : $3^{4x^2} = e^t$

⑥ Evaluate $\int_0^1 (\log x)^5 dx$

Hint put : $\log x = -t$ Ans -120

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Beta function :⇒

Beta function $\beta(m,n)$ is defined as

$$\boxed{\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx}, \quad m>0,\ n>0$$

$\beta(m,n)$ is also known as Euler's Integral of first kind.

OR $\beta(m+1, n+1) = \int_0^1 x^m (1-x)^n\,dx$, $m>0, n>0$

★ Trigonometric form of $\beta(m,n)$

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta\,d\theta, \quad \begin{matrix} m>0 \\ n>0 \end{matrix}$$

Proof :

By defn of Beta function.

$$\beta(m,n) = \int_0^1 x^{m-1}\cdot(1-x)^{n-1}\,dx \quad \text{——(1)}$$

Put $x = \sin^2\theta \Rightarrow dx = 2\sin\theta\cdot\cos\theta\cdot d\theta$

and as $x=0 \Rightarrow \sin^2\theta = 0$ i.e. $\theta = \sin^{-1}0 = 0$

as $x=1 \Rightarrow \sin^2\theta = 1$ i.e. $\theta = \sin^{-1}(1) = \pi/2$

$\therefore$ (1) $\Rightarrow$

$$\beta(m,n) = \int_0^{\pi/2} (\sin^2\theta)^{m-1}(1-\sin^2\theta)^{n-1}\cdot 2\sin\theta\cdot\cos\theta\,d\theta$$

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$$= 2\int_0^{\pi/2} \sin^{2m-2}\theta \cdot \cos^{2n-2}\theta \cdot \sin\theta\cdot\cos\theta\, d\theta$$

$$\boxed{\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta\, d\theta}$$

★ Corollary.

Put $m = \frac{p+1}{2}$, $n = \frac{q+1}{2}$

$$\therefore \boxed{\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2\int_0^{\pi/2} \sin^{p}\theta \cdot \cos^{q}\theta\, d\theta}$$

★ Properties of Beta function :

① $\beta(m,n) = \beta(n,m)$

② Proof : By defⁿ of Beta function

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$$

put $1-x = t \Rightarrow dx = -dt$

and as $x = 0 \Rightarrow t = 1$

as $x = 1 \Rightarrow t = 0$

$$\therefore \beta(m,n) = \int_{t=1}^{0} (1-t)^{m-1} \cdot t^{n-1}\, (-dt)$$

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$$= -\int_1^0 t^{n-1} \cdot (1-t)^{m-1} \cdot dt$$

$$\beta(m,n) = \int_0^1 t^{n-1} \cdot (1-t)^{m-1} \, dt$$

$$= \beta(n,m)$$

Hence $\boxed{\beta(m,n) = \beta(n,m)}$

i.e. In β Beta function m & n are interchangeable.

★ Relation betn Gamma and Beta function

$$\beta(m,n) = \frac{\lceil m \, \lceil n}{\lceil m+n}$$

$$\int_0^\infty e^{-x} x^{m-1} dx \cdot \int_0^\infty e^{-x} x^{n-1} dx$$

$$\int_0^\infty e^{-x} x^{m+n-1} dx$$

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Ex: Find the values of

(i) $\beta\left(\frac{3}{2}, \frac{1}{2}\right)$ (ii) $\beta\left(\frac{4}{3}, \frac{5}{3}\right)$

Soln (i)

$$\beta\left(\tfrac{3}{2}, \tfrac{1}{2}\right) = \frac{\lceil 3/2 \cdot \lceil 1/2}{\lceil 3/2 + 1/2}$$

$$= \frac{\frac{1}{2}\lceil 1/2 \cdot \lceil 1/2}{\lceil 2} \qquad \because \lceil 3/2 = \lceil 1/2 + 1 = \tfrac{1}{2}\lceil 1/2$$

$$= \frac{1}{2}\sqrt{\pi}\cdot\sqrt{\pi} \qquad \because \lceil 2 = 1! = 1$$

$$\therefore \boxed{\beta\left(\tfrac{3}{2}, \tfrac{1}{2}\right) = \pi/2}$$

(ii) $\beta\left(\frac{4}{3}, \frac{5}{3}\right) = \frac{\lceil 4/3 \cdot \lceil 5/3}{\lceil \frac{4}{3} + 5/3}$

$$= \frac{\lceil 4/3 \cdot \lceil 5/3}{\lceil 3}$$

$$= \frac{1}{2!}\lceil 1/3 + 1 \quad \lceil \tfrac{2}{3} + 1$$

$$= \frac{1}{2} \cdot \frac{1}{3}\lceil 1/3 \cdot \frac{2}{3}\lceil 2/3$$

$$= \frac{1}{9}\lceil 1/3 \cdot \lceil 2/3$$

$$= \frac{1}{9}\lceil 1/3 \cdot \lceil 1 - 1/3 \qquad \because \lceil n \lceil 1-n = \frac{\pi}{\sin n\pi}$$

$$= \frac{1}{9}\frac{\pi}{\sin \pi/3} = \frac{1}{9}\frac{\pi}{(\sqrt{3}/2)}$$

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$$\therefore \boxed{\beta\left(\frac{4}{3}, \frac{5}{3}\right) = \frac{2\pi}{9\sqrt{3}}}$$

Ex: If $\beta(k, 3) = \frac{1}{60}$. find the ~~value of~~ +ve integer k

Soln: We know that, By defn of β-function

$$\frac{1}{60} = \beta(k,3)$$

$$= \frac{\Gamma k \cdot \Gamma 3}{\Gamma 3+k}$$

$$= \frac{2 \cdot \Gamma k}{\Gamma(2+k)+1} \qquad \because \Gamma 3 = 2! = 2$$

$$= \frac{2\,\Gamma k}{(2+k)\,\Gamma 2+k} \qquad \because \Gamma n+1 = n\,\Gamma n$$

$$= \frac{2\,\Gamma k}{(2+k)(1+k)\,\Gamma 1+k} \qquad \because \Gamma 2+k = \Gamma 1+k+1 = (1+k)\Gamma 1+k$$

$$= \frac{2\,\Gamma k}{(2+k)(1+k)\,k\,\Gamma k}$$

$$\Rightarrow \frac{1}{60} = \frac{2}{k(1+k)(2+k)}$$

$$\Rightarrow k(1+k)(2+k) = 120$$

$$\Rightarrow k^3 + 3k^2 + 2k = 120$$

$$\Rightarrow k^3 + 3k^2 + 2k - 120 = 0$$

$$\therefore k = 4, -3.5, -3.5$$

Req^d $\boxed{k = 4}$ which is +ve integer.

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Ex: Evaluate $\int_0^2 x\,(8-x^3)^{1/3}\,dx$

Soln: Let

$$I = \int_0^2 x\,(8-x^3)^{1/3}\,dx$$

$$= \int_0^2 x\,(8)^{1/3}\cdot(1-x^3/8)^{1/3}\,dx$$

$$= 2\int_0^2 x\cdot(1-x^3/8)^{1/3}\,dx$$

put $\frac{x^3}{8} = t \Rightarrow x^3 = 8t$

$\Rightarrow dx = 2t^{1/3}\,dt$

as $x=0 \Rightarrow t=0$

$x=2 \Rightarrow t=1$

$\therefore dx = \frac{2}{3}t^{-2/3}\,dt$

$$\therefore I = 2\int_0^1 (8t)^{1/3}\,(1-t)^{1/3}\cdot\frac{2}{3}t^{-2/3}\,dt$$

$$= \frac{8}{3}\int_0^1 t^{-1/3}\,(1-t)^{1/3}\,dt$$

$$= \frac{8}{3}\int_0^1 t^{2/3-1}\,(1-t)^{4/3-1}\,dt$$

$$I = \frac{8}{3}\beta\left(\frac{2}{3},\frac{4}{3}\right) \quad \because \text{by defn of } \beta(m,n)$$

$$= \frac{8}{3}\,\frac{\Gamma 2/3\cdot\Gamma 4/3}{\Gamma\frac{2}{3}+4/3}$$

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$$= \frac{8}{3} \frac{\Gamma(2/3)\,\Gamma(\frac{1}{3}+1)}{\Gamma 2}$$

$$= \frac{8}{3} \cdot \frac{1}{3}\,\Gamma(1/3)\cdot\Gamma(2/3)$$

$$= \frac{8}{9} \cdot \Gamma(1/3)\,\Gamma(1-1/3)$$

$$= \frac{8}{9} \frac{\pi}{\sin \pi/3} \qquad \because \Gamma n\,\Gamma(1-n) = \frac{\pi}{\sin n\pi}$$

$$= \frac{8}{9} \cdot \frac{2}{\sqrt{3}}\,\pi \qquad \because \sin \pi/3 = \frac{\sqrt{3}}{2}$$

$$\therefore \int_0^2 x\,(8-x^3)^{1/3}\,dx = \frac{16\pi}{9\sqrt{3}}$$



Ex. **Evaluate.** $\int_0^{2a} x\sqrt{2ax-x^2}\,dx$

Soln.

Let $I = \int_0^{2a} x\sqrt{2ax-x^2}\,dx$

$$= \int_0^{2a} x\sqrt{2ax}\sqrt{1-\frac{x^2}{2ax}}\cdot dx$$

$$= \sqrt{2a}\int_0^{2a} x^{3/2}\cdot(1-x/2a)^{1/2}\,dx$$

put $\frac{x}{2a} = t \Rightarrow dx = 2a\,dt$

as $x=0 \Rightarrow t=0$

as $x=2a \Rightarrow t=1$

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$$\therefore I = \sqrt{2a} \int_0^{2a} (2at)^{3/2} (1-t)^{1/2} \cdot 2a\, dt$$

$$= (2a)^{1/2+3/2+1} \int_0^1 t^{3/2} (1-t)^{1/2}\, dt$$

$$= 8a^3 \int_0^1 t^{5/2-1} \cdot (1-t)^{3/2-1}\, dt$$

$$= 8a^3 \cdot \beta\left(\frac{5}{2}, \frac{3}{2}\right) \quad \because \text{by defn.}$$

$$= 8a^3 \frac{\Gamma 5/2 \; \Gamma 3/2}{\Gamma \frac{5}{2}+3/2}$$

$(\because \Gamma n+1 = n\,\Gamma n)$

$$= 8a^3 \cdot \frac{(3/2 \; 1/2 \; \Gamma 1/2) \; (\frac{1}{2} \Gamma 1/2)}{\Gamma 4}$$

$\because \Gamma 1/2 = \sqrt{\pi}$

$$= \frac{8a^3}{8} \frac{(3)\pi}{3!}$$

$$= \frac{a^3 \pi}{2}$$

$$\therefore \int_0^{2a} x \sqrt{2ax - x^2}\, dx = \frac{a^3 \pi}{2}$$

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Ex: Prove that $\int_a^b (x-a)^m (b-x)^n \, dx = (b-a)^{m+n+1} \cdot \beta(m+1, n+1)$

Soln:

Let $I = \int_{x=a}^{b} (x-a)^m \cdot (b-x)^n \, dx$ ——— (1)

put $(x-a) = (b-a) \cdot t \Rightarrow dx = (b-a)\, dt$

and as $x = a \Rightarrow t = 0$

$x = b \Rightarrow t = 1$

$\Rightarrow$ put $\boxed{t = \left(\frac{x-a}{b-a}\right)}$

$(1) \Rightarrow I = \int_0^1 [(b-a)\cdot t]^m \left(b - \{(b-a)t + a\}\right)^n \cdot (b-a)\, dt$

$= \int_0^1 (b-a)^{m+1} \cdot t^m \cdot [(b-a)(1-t)]^n \cdot dt$

$= (b-a)^{m+n+1} \int_0^1 t^m \cdot (1-t)^n \cdot dt$

$= (b-a)^{m+n+1} \int_0^1 t^{(m+1)-1} \cdot (1-t)^{(n+1)-1} \, dt$

$= (b-a)^{m+n+1} \cdot \beta(m+1, n+1)$

H.P.

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OR $\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = 2\int_0^{\pi/2} \sin^p\theta \cdot \cos^q\theta \, d\theta$

Trigonometric form of β-function

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta \, d\theta$$

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Problems based on Trigonometric form of β-function

OR

Ex: Evaluate $\int_0^{\pi/2} \sqrt{\cot\theta}\, d\theta$

Soln:

Let $I = \int_0^{\pi/2} (\cot\theta)^{1/2} \, d\theta = \int_0^{\pi/2} \frac{\cos^{1/2}\theta}{\sin^{1/2}\theta} \, d\theta$

$= \int_0^{\pi/2} \sin^{-1/2}\theta \cdot \cos^{1/2}\theta \, d\theta$

$= \int_0^{\pi/2} \sin^{\frac{1}{2}-1}\theta \cdot \cos^{3/2-1}\theta \, d\theta$

$= \frac{\beta\left(\frac{1}{2}, \frac{3}{2}\right)}{2}$

$\because \beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta$

$\Rightarrow \frac{\beta(m,n)}{2} = \int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta \, d\theta$

$= \int_0^{\pi/2} \sin^{2(1/4)-1}\theta \cdot \sin^{2(3/4)-1}\theta \, d\theta$

$= \frac{1}{2}\beta\left(\frac{1}{4}, \frac{3}{4}\right)$

OR $2\int_0^{\pi/2} \sin^p\theta \cos^q\theta \, d\theta = \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

$= \frac{1}{2}\Gamma\frac{1}{4} \cdot \Gamma\frac{3}{4}$

$= \frac{1}{2}\Gamma\frac{1}{4}\,\Gamma\left(1-\frac{1}{4}\right)$

$= \frac{1}{2}\cdot\frac{\pi}{\sin\pi/4}$ $\quad \because \Gamma n\,\Gamma(1-n) = \pi/\sin n\pi$

$= \frac{\pi}{\sqrt{2}}$ Ans.

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Ex: Duplication formula Show that $\beta(m,m) = \frac{\sqrt{\pi}\,\sqrt{m}}{2^{2m-1}\cdot\sqrt{m+1/2}}$

Show that $\overline{m}\ \overline{m+\frac{1}{2}} = \frac{\sqrt{\pi}\cdot\overline{2m}}{2^{2m-1}}$

OR $\overline{2m} = \frac{2^{2m-1}}{\sqrt{\pi}}\cdot\overline{m}\ \overline{m+\frac{1}{2}}$.

Soln: We know that

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\cdot\cos^{2n-1}\theta\, d\theta$$

put $n=m$. we get

$$\beta(m,m) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\cdot\cos^{2m-1}\theta\cdot d\theta$$

$$= 2\int_0^{\pi/2} (\sin\theta\cdot\cos\theta)^{2m-1}\cdot d\theta$$

$$= 2\int_0^{\pi/2} \left(\frac{\sin 2\theta}{2}\right)^{2m-1} d\theta \quad \because \sin 2\theta = 2\sin\theta\cdot\cos\theta$$

$$\beta(m,n) = \frac{2}{2^{2m-1}}\int_0^{\pi/2} (\sin 2\theta)^{2m-1}\cdot d\theta \quad \text{---(1)}$$

put $2\theta = t \Rightarrow d\theta = dt/2$

as $\theta = 0 \Rightarrow t = 0$

$\theta = \pi/2 \Rightarrow t = \pi$

$$\therefore \beta(m,n) = \frac{2}{2^{2m-1}}\cdot\int_0^{\pi} (\sin t)^{2m-1}\cdot\frac{dt}{2}$$

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$$= \frac{1}{2^{2m-1}} \cdot \int_0^{\pi} (\sin t)^{2m-1} dt$$

$$= \frac{1}{2^{2m-1}} \cdot \left[2 \int_0^{\pi/2} (\sin t)^{2m-1} \cdot dt \right] \quad \because \sin(\pi/2 - t) = \sin t.$$

we know

$$\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$$

if $f(2a-x) = f(x)$

$$= \frac{1}{2^{2m-1}} \left[2 \int_0^{\pi/2} \sin^{2m-1} t \cdot \cdot dt \right]$$

$$= \frac{1}{2^{2m-1}} \left[2 \int_0^{\pi/2} \cdot \sin^{2m-1} t \cdot \cos^{2(1/2)-1} t \, dt \right] \quad \because \cos^0 t = 1 = \Rightarrow \cos^{2(1/2)-1} t = 1$$

$$= \frac{1}{2^{2m-1}} \cdot \beta(m, 1/2)$$

$$= \frac{1}{2^{2m-1}} \cdot \frac{\Gamma m \, \Gamma 1/2}{\Gamma m+1/2}$$

$$\boxed{\beta(m,m) = \frac{1}{2^{2m-1}} \cdot \frac{\sqrt{\pi} \cdot \Gamma m}{\Gamma m+1/2}}$$

$$\Rightarrow \frac{\Gamma m \, \Gamma m}{\Gamma m+m} = \frac{\sqrt{\pi} \cdot \Gamma m}{2^{2m-1} \cdot \Gamma m+1/2}$$

$$\Rightarrow \boxed{\Gamma m \cdot \Gamma m+1/2 = \frac{\sqrt{\pi} \cdot \Gamma 2m}{2^{2m-1}}}$$

Hence $$\Gamma 2m = \frac{2^{2m-1}}{\sqrt{\pi}} \cdot \Gamma m \, \Gamma m+1/2$$

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Ex: Evaluate $\int_0^{2\pi} \sin^2\theta\,(1+\cos\theta)^4\,d\theta$

Soln:

Let $I = \int_0^{2\pi} \sin^2\theta \cdot (1+\cos\theta)^4\,d\theta$ ——— (1)

$$= \int_0^{2\pi} \cdot \left(2\sin\theta/2 \cdot \cos\theta/2\right)^2 \cdot \left[2\cos^2\theta/2\right]^4 d\theta$$

$$= 2^6 \cdot \int_0^{2\pi} \sin^2\theta/2 \cdot \cos^{10}\theta/2\,d\theta \quad \text{——— (1)}$$

put $\theta/2 = t \quad \therefore d\theta = 2\,dt$

as $\theta = 0 \Rightarrow t = 0$

as $\theta = 2\pi \Rightarrow t = \pi$

$\therefore$ (1) $\Rightarrow$ $I = 2^6 \cdot \int_0^{\pi} \sin^2 t \cdot \cos^{10} t\,(2\,dt)$

$$= 2^7 \int_0^{\pi} \sin^2 t \cdot \cos^{10} t\,dt$$

$$= 2^7 \cdot 2 \int_0^{\pi/2} \sin^2 t \cdot \cos^{10} t\,dt$$

$\because \int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$ if $f(2a-x) = f(x)$

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$$= 2^7 \left[2 \int_0^{\pi/2} \sin^{2\left(\frac{3}{2}\right)-1} t \cdot \cos^{2\left(\frac{11}{2}\right)-1} t \, dt \right]$$

$$= 2^7 \cdot \beta\left(\tfrac{3}{2}, \tfrac{11}{2}\right)$$

$$= 2^7 \cdot \frac{\lceil 3/2 \cdot \lceil 11/2}{\lceil \frac{3}{2} + 11/2}$$

$$= 2^7 \cdot \frac{1/2\left(\lceil 1/2\right) \cdot 9/2 \cdot 7/2 \cdot 5/2 \cdot 3/2 \cdot 1/2\left(\lceil 1/2\right)}{\lceil 7}$$

$$= \frac{2^7}{6!} \cdot \pi \cdot \frac{(9 \times 7 \times 5 \times 3)}{2^6}$$

$$= \frac{2}{6!} \pi \cdot (945)$$

$$= \frac{\pi \cdot (945)}{360}$$

$$= \frac{63 \cdot \pi}{24} = \frac{21\pi}{8}$$

$$\boxed{\int_0^{2\pi} \sin^2\theta \cdot (1+\cos\theta)^4 \, d\theta = \frac{21\pi}{8}}$$

$$\begin{array}{r} 15 \\ \times 7 \\ \hline 105 \\ \times 9 \\ \hline 945 \end{array}$$

$6! = 6 \times 5 \times 4 \times 3 \times 2$

$$\begin{array}{r} 21 \\ \times 15 \\ \hline 6 \end{array}$$

$$\begin{array}{r} 24 \\ \times 15 \\ \hline 6 \end{array}$$

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Some standard Results of β-function as improper Integral

Ex: Show that

$$\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx$$

$$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$$

Soln

let $$I = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx \quad \text{---(1)}$$

Put $x = \tan^2\theta \Rightarrow dx = 2\tan\theta \cdot \sec^2\theta\, d\theta$

as $x = 0 \Rightarrow \tan\theta = 0 \therefore \theta = 0$

as $x = \infty \Rightarrow \tan\theta = \infty \therefore \theta = \tan^{-1}\infty = \pi/2$

$$\text{(1)} \Rightarrow I = \int_0^{\pi/2} \frac{(\tan^2\theta)^{m-1}}{(1+\tan^2\theta)^{m+n}} \cdot 2\tan\theta \cdot \sec^2\theta\, d\theta$$

$$= 2\int_0^{\pi/2} \frac{\tan^{2m-1}\theta \cdot \sec^2\theta}{(\sec\theta)^{2m+2n}}\,d\theta$$

$$= 2\int_0^{\pi/2} \tan^{2m+1}\theta \cdot \sec^{2(m+n+1)}\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \frac{\sin^{2m-1}\theta}{\cos^{2m-1}\theta} \cdot \frac{1}{\cos^{-2m}\cdot\cos^{2-2n}\theta}\,d\theta$$

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$$= 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-2}\theta \cdot \cos\theta \, d\theta$$

$$= 2\int_0^{\pi/2} \sin^{2m-1}\theta \cdot \cos^{2n-1}\theta \, d\theta$$

$$= \beta(m,n)$$

Hence $$\boxed{\beta(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx}$$

Interchanging m and n, we get

$$\beta(n,m) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{n+m}} \, dx = \beta(m,n)$$

Ex: Show that

$$\beta(m,n) = \int_0^{1} \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} \, dx$$

Soln We know that

$$\beta(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx \quad \text{---(1)}$$

Now,

$$= \int_0^{1} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx + \int_1^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx$$

---(1)

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let us consider

$$I = \int_1^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx$$

put $x = \frac{1}{y} \Rightarrow dx = \frac{-1}{y^2} \cdot dy$

as $x = 1 \Rightarrow y = 1$

$x = \infty \Rightarrow y = 0$

$$\therefore I = \int_1^0 \frac{(1/y)^{m-1}}{(1+1/y)^{m+n}} \cdot \left(\frac{-1}{y^2}\right) dy$$

$$= \int_0^1 \frac{\left(\frac{1}{y^{m-1}} \cdot \frac{1}{y^2}\right) \cdot dy}{\left(\frac{1+y}{y}\right)^{m+n}}$$

$\because \int_a^b f(x)\,dx = \int_a^a f(x)\,dx$

$$= \int_0^1 \frac{y^{m+n}}{y^{m+1} \cdot (1+y)^{m+n}}\, dy$$

$$= \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}}\, dy$$

$\because \int f(x)\,dx = \int f(y)\,dy$

$$= \int_0^1 \frac{x^{n-1}}{(1+x)^{m+n}}\, dx$$

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$\therefore$ eqn (1) $\Rightarrow$ $\beta(m,n) = \int_0^{\infty} \frac{x^{m+1}}{(1+x)^{m+n}}\,dx + \int_0^{1} \frac{x^{n-1}}{(1+x)^{n+1}}\,dx$

$$\boxed{\beta(m,n) = \int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}\,dx}$$ H.P.

(For Reader only) $\longleftrightarrow$

Ex: Show that $\beta(n, 1-n) = \frac{\pi}{\sin n\pi}$

OR

$$\overline{n}\,\overline{1-n} = \frac{\pi}{\sin n\pi}$$

Soln: We know that

$$\frac{\overline{m}\,\overline{n}}{\overline{m+n}} = \beta(m,n) \quad \leftarrow (1)$$

Replace m by $1-n$

$$\therefore \overline{n}\,\overline{1-n} = \beta(n, 1-n)$$

$$= \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{n+(1-n)}}\,dx$$

$$= \int_0^{\infty} \frac{x^{n-1}}{(1+x)}\,dx \qquad \text{for } 0<n<1$$

$$= \int_0^{\infty} x^{n-1} \cdot F(x) \cdot dx$$

(By standard Result of Residue thm)

$\Rightarrow \overline{n}\,\overline{1-n} = \frac{\pi}{\sin n\pi} \cdot \sum$ Residue of $(-z)^{n-1} \cdot F(z)$ at each pole $\quad \leftarrow (1)$

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Here $f(x) = \frac{1}{1+x}$

i.e. $F(z) = \frac{1}{1+z}$

and the pole of $f(z)$ is $z = -1$ (simple pole)

$$\left[\begin{array}{l}\text{Residue of } (-z)^{n-1} f(z) \\ \text{at } z=-1\end{array}\right] = \lim_{z \to -1} (z+1) \cdot (-z)^{n-1} \cdot F(z)$$

$$= \lim_{z \to -1} (z+1) \cdot (-z)^{n-1} \cdot \frac{1}{(1+z)}$$

$$= (1)^{n-1}$$

$$= 1.$$

and Hence

$$\textcircled{1} \Rightarrow \Gamma n \, \Gamma 1-n = \frac{\pi}{\sin n\pi} \cdot (1) = \frac{\pi}{\sin n\pi}.$$

$$\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma m+n}$$

$$\Rightarrow \Gamma m \, \Gamma n = \Gamma m+n \cdot \beta(m,n)$$

put $m = 1-n$

$$\Gamma n \, \Gamma 1-n = \Gamma 1 \, \beta(1-n, n)$$

$$= \int_0^\infty \frac{x^{n-1}}{(1+x)^1} \, dx = \frac{\pi}{\sin n\pi}$$

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~~How~~ Hw

Ex: ① Express $\int_0^1 x^m (1-x^n)^p \, dx$ in terms of Beta function and Hence evaluate $\int_0^1 x^5 (1-x^3)^{10} \, dx$.

Ans (i) $\frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right)$

(ii) $\frac{2}{3 \times (10!)}$

② Evaluate $\int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} \, dx = \frac{1}{a^{m+n}} \int_0^\infty \frac{x^{m-1}}{\left(1+\frac{bx}{a}\right)^{m+n}}$ Ans $\Rightarrow \frac{\beta(m,n)}{a^m b^n}$

Put $\frac{bx}{a} = t$.

③ Evaluate $\int_0^\pi \sin^5 x \cdot (1-\cos x)^3 \, dx$ Ans $32/21$

④ prove that $\int_0^{\pi/2} \sqrt{\tan\theta} \, d\theta = \frac{\pi}{\sqrt{2}}$

⑤ Given that $\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} \, dx = \frac{\pi}{\sin n\pi}$

Show that $\overline{n} \, \overline{1-n} = \frac{\pi}{\sin n\pi}$.

Hint: $\overline{m} \, \overline{n} = \overline{m+n} \cdot \beta(m,n)$

also prove that $\overline{1/2} = \sqrt{\pi}$.

⑥ Show that $\beta(m,n) = \beta(m, n+1) + \beta(m+1, n)$

⑦ Show that $\beta(m+1, n) = \frac{m}{m+n} \beta(m,n)$

⑧ Evaluate $\int_0^{2a} x \sqrt{2ax - x^2} \, dx$ Ans $\frac{a^3\pi}{2}$

⑨ prove that $\int_0^{\pi/2} \sin^{m-1}\theta \cdot \cos^{n-1}\theta \, d\theta = \frac{\overline{m/2} \cdot \overline{n/2}}{2 \, \overline{\frac{m+n}{2}}}$

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Differentiation under Integral sign :⇒

Let $f(x, \alpha)$ be the function of two variable x and α where α is parameter.

Then

$$\int_a^b f(x, \alpha)\, dx = F(\alpha) \quad \text{—(1)}$$

Then derivative of $F(\alpha)$ w.r.to. to α is given by following rules

(i) Leibnitz Rule - I

If Limits of integrations are constant. (not depend on α and x.

then

$$\boxed{\frac{d}{d\alpha} F(\alpha) = \int_a^b \frac{\partial}{\partial \alpha} f(x, \alpha)\, dx.}$$

(ii) Leibnitz rule - II

If limits of integrations are function of parameter α.

i.e. $$F(\alpha) = \int_{\phi(\alpha)}^{\psi(\alpha)} f(x, \alpha)\, dx$$

Then

$$\frac{d F(\alpha)}{d\alpha} = \int_{\phi(\alpha)}^{\psi(\alpha)} \frac{\partial}{\partial \alpha} f(x, \alpha)\, dx + \frac{d\psi(\alpha)}{d\alpha} \cdot f(\psi(\alpha), \alpha) - \frac{d\phi(\alpha)}{d\alpha} \cdot f(\phi(\alpha), \alpha).$$

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Ex ① Evaluate $\int_0^\infty e^{-\left(x^2+\frac{a^2}{x^2}\right)} \cdot dx$

by differentiating under integral sign.

given that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

Sol: let

$$F(a) = \int_0^\infty e^{-(x^2+a^2/x^2)} dx. \quad \text{---} ①$$

Diff. w.r.to a, we get

$$\frac{dF(a)}{da} = \frac{d}{da}\left(\int_0^\infty e^{-(x^2+a^2/x^2)} \cdot dx\right)$$

$$= \int_0^\infty \frac{\partial}{\partial a} e^{-(x^2+a^2/x^2)} \cdot dx \qquad (\because \text{by Leibnitz rule } ①)$$

$$= \int_0^\infty e^{-(x^2+a^2/x^2)} \left(\frac{-2a}{x^2}\right) dx$$

$$\frac{dF(a)}{da} = \int_0^\infty e^{-(x^2+a^2/x^2)} \cdot \left(\frac{-2a}{x^2}\right) dx \quad \text{---} ②$$

Put $a/x = t \Rightarrow x = a/t$

$$\therefore dx = \frac{-a}{t^2} dt$$

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As $x = 0 \Rightarrow t = \infty$

$x = \infty \Rightarrow t = 0$

$$\textcircled{2} \Rightarrow \therefore \frac{dF(a)}{da} = \int_{\infty}^{0} e^{-\left(\frac{a^2}{t^2} + t^2\right)} \cdot -2a\left(\frac{t^2}{a^2}\right) \cdot \left(\frac{-a}{t^2}\right) dt$$

$$= 2a \int_{\infty}^{0} e^{-(t^2 + a^2/t^2)} \cdot dt$$

$$= -2 \int_{0}^{\infty} e^{-(t^2 + a^2/t^2)} \, dt \quad \left\{ \int_a^b = -\int_b^a \right\}$$

$$\frac{dF(a)}{da} = -2F(a) \quad (\because \text{By eqn } \textcircled{1})$$

$$\Rightarrow \frac{dF(a)}{F(a)} = -2 \, da \quad \text{(separating variables)}$$

on integrating we get

$$\log F(a) = -2a + \log c$$

$$\therefore \log F(a) - \log c = -2a$$

$$\Rightarrow \log\left(\frac{F(a)}{c}\right) = -2a$$

$$\Rightarrow \boxed{F(a) = c \cdot e^{-2a}} \quad \textcircled{3}$$

$$\Rightarrow \int_0^{\infty} e^{-(x^2 + a^2/x^2)} \, dx = c\, e^{-2a} \quad \textcircled{3}$$

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To find Const: C

put $q=0$ in eqn (1) we get

$$F(0) = \int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \qquad (\because \text{given condn})$$

also eqn (3) become

$$F(0) = C\,e^{0} = C$$

$$\therefore C = \frac{\sqrt{\pi}}{2}$$

$\therefore$ Req^n soln is

$$F(q) = \frac{\sqrt{\pi}}{2} e^{-2q}$$

i.e.

$$\int_0^\infty e^{-(x^2 + q^2/x^2)}\,dx = \frac{\sqrt{\pi}}{2} e^{-2q}$$

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Note:⇒ $\int e^{ax}\cos bx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx]$

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Ex: • Evaluate $\int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \cdot \sec x}\, dx$, $\alpha > 0$

Using differentiation under integral. also find $F(2)$.

OR

• Show that.

$$\int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \cdot \sec x}\, dx = \frac{1}{2}\log\left(\frac{\alpha^2+1}{2}\right).$$

Soln: Let

$$F(\alpha) = \int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \sec x}\, dx \quad \text{---(1)}$$

diff. w.r.to α we get

$$\frac{dF(\alpha)}{d\alpha} = \frac{d}{d\alpha}\int_0^\infty \left(\frac{e^{-x} - e^{-\alpha x}}{x \sec x}\right) dx$$

$$= -\int_0^\infty \frac{\partial}{\partial \alpha}\left(\frac{e^{-x} - e^{-\alpha x}}{x \sec x}\right) \cdot dx \quad \because \left[\text{by leibnitz rule } \textcircled{B}\right.$$

$$= \int_0^\infty \frac{1}{x \sec x} \cdot \left(0 + x e^{-\alpha x}\right) \cdot dx$$

$$= \int_0^\infty \frac{e^{-\alpha x}}{\sec x}\, dx$$

$$\frac{dF(\alpha)}{d\alpha} = \int_0^\infty e^{-\alpha x} \cdot \cos x\, dx$$

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$$\therefore \frac{dF(\alpha)}{d\alpha} = \left\{ \frac{e^{-\alpha x}}{(\alpha^2+1^2)} \cdot [-\alpha \cos x + 2\sin x] \right\}_0^\infty$$

$$\because \int e^{ax} \cos bx = \frac{e^{ax}}{a^2+b^2} (a\cos bx + b \sin bx$$

$$= \left[0 - \frac{1}{\alpha^2+1} \cdot (-\alpha) \right]$$

$$\frac{dF(\alpha)}{d\alpha} = \frac{\alpha}{\alpha^2+1}$$

Integrating on b.s. we get

$$F(\alpha) = \int \frac{\alpha}{\alpha^2+1} d\alpha + C$$

$$\boxed{F(\alpha) = \frac{1}{2} \log(\alpha^2+1) + C} \quad \longrightarrow (2)$$

Put $\alpha = 1$ in eqn (1) & (2). we get

From (1).

$$F(1) = 0$$

&

$$(2) \Rightarrow F(1) = \frac{1}{2} \log(2) + C$$

ie $$0 = \frac{1}{2} \log 2 + C$$

$$\Rightarrow C = -\frac{1}{2} \log 2$$

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∴ Reqd. soln. is

$$F(\alpha) = \frac{1}{2}\log(\alpha^2+1) - \frac{1}{2}\log 2$$

$$= \frac{1}{2}\log\left(\frac{\alpha^2+1}{2}\right)$$

(or)

$$F(\alpha) = \int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \sec x}\, dx = \frac{1}{2}\log\left(\frac{\alpha^2+1}{2}\right)$$

Now put $\alpha = 2$

$$\Rightarrow F(2) = \frac{1}{2}\log\left(\frac{5}{2}\right) = \frac{1}{2}\log(5/2).$$



$$\Rightarrow \text{GP} \qquad F(\alpha) = \frac{1}{2}\log(\alpha^2+1) + C$$

$$\int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \sec x}\, dx = \frac{1}{2}\log(\alpha^2+1) + C \qquad \text{---(1)}$$

To find c : put $\alpha = 1$ $\Rightarrow$ $0 = \frac{1}{2}\log(2) + C$

$$\Rightarrow C = -\frac{1}{2}\log 2$$

∴ Reqd soln is $\int_0^\infty \frac{e^{-x} - e^{-\alpha x}}{x \sec x}\, dx = \frac{1}{2}\log(\alpha^2+1) - \frac{1}{2}\log 2$

$$= \log\left(\frac{\alpha^2+1}{2}\right)^{1/2}$$

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Ex: By differentiating under the integral sign.

Evaluate $\int_0^{\pi} \frac{\log(1+\sin\alpha \cdot \cos x)}{\cos x}\, dx$

Sol: Let

$$F(\alpha) = \int_0^{\pi} \frac{\log(1+\sin\alpha \cdot \cos x)}{\cos x}\, dx \quad \text{---(1)}$$

Diff w.r.to. α, (by Leibnitz rule (1))

$$\frac{d\,F(\alpha)}{d\alpha} = \frac{d}{d\alpha} \int_0^{\pi} \frac{\log(1+\sin\alpha \cdot \cos x)}{\cos x}\, dx$$

$$= \int_0^{\pi} \frac{1}{\cos x} \left(\frac{\partial}{\partial \alpha} \log(1+\sin\alpha \cdot \cos x) \right) dx$$

$$= \int_0^{\pi} \frac{1}{\cos x} \cdot \frac{\cos x \cdot \cos x}{1+\sin\alpha \cdot \cos x}\, dx$$

$$= \int_0^{\pi} \frac{\cos\alpha}{1+\sin\alpha \cdot \cos x}\, dx$$

Put $\cos\alpha = a$ and $\sin\alpha = b$

$$\therefore \frac{d\,F(\alpha)}{d\alpha} = \int_0^{\pi} \frac{a}{1+b\cos x}\, dx \quad \text{---(2)}$$

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$$\Rightarrow \frac{dF(\alpha)}{d\alpha} = a\int_0^{\pi} \frac{1}{1+b\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right)}\,dx$$

$$\because \left(\cos\theta = \frac{1-\tan^2\theta/2}{1+\tan^2\theta/2}\right.$$

put $\tan x/2 = t$

$$\Rightarrow \frac{1}{2}\sec^2 x/2\, dx = dt \quad \Rightarrow dx = \frac{2\,dt}{\sec^2 x/2}$$

$$= \frac{2\,dt}{1+\tan^2 x/2} = \frac{2dt}{1+t^2}$$

as $x = 0 \Rightarrow t = 0$

$x = \pi \Rightarrow t = \infty$

$$\therefore \frac{dF(\alpha)}{d\alpha} = a\int_0^{\infty} \frac{\left(\frac{2}{1+t^2}\right)\cdot dt}{1+b\left(\frac{1-t^2}{1+t^2}\right)}$$

$$= a\int_0^{\infty} \frac{2}{(1+t^2)+b(1-t^2)}\,dt$$

$$= a\int_0^{\infty} \frac{2}{(1+b)+(1-b)t^2}\,dt$$

$$= \frac{2a}{(1-b)}\int_0^{\infty} \frac{1}{t^2+\left(\frac{1+b}{1-b}\right)}\,dt$$

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$$= \frac{2q}{(1-b)} \int_0^\infty \frac{1}{t^2 + \left(\sqrt{\frac{1+b}{1-b}}\right)^2} dt$$

$$= \frac{2q}{(1-b)} \cdot \left[\frac{1}{\sqrt{\frac{1+b}{1-b}}} \cdot \tan^{-1}\left(\frac{t}{\sqrt{\frac{1+b}{1-b}}} \right) \right]_0^\infty$$

$\because \int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1}(\frac{x}{a})$

$$= \frac{2q}{(1-b)} \cdot \frac{1}{\sqrt{\frac{1+b}{1-b}}} \cdot \left(\pi/2 - 0 \right)$$

$$= \frac{2q}{\sqrt{(1-b)(1+b)}} \quad (\pi/2)$$

$$\frac{dF(\alpha)}{d\alpha} = \frac{2q\pi}{\sqrt{1-b^2}}$$

$$\frac{dF(\alpha)}{d\alpha} = \frac{\pi \cos\alpha}{\sqrt{1-\sin^2\alpha}} = \frac{\pi \cdot \cos\alpha}{\cos\alpha}$$

$\because a = \cos\alpha$
$b = \sin\alpha$

$$\therefore \boxed{\frac{dF(\alpha)}{d\alpha} = \pi}$$

integrating w.r.to α, we get

$$F(\alpha) = \pi\alpha + C \qquad \text{---------- (2)}$$

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To find c

Put $\alpha = 0$ in eqn (1) and (2).

(1) $\Rightarrow$ $F(0) = \int_0^{\pi} \frac{\log(1+0)}{\cos x}\, dx = 0$.

(2) $\Rightarrow$ $F(0) = c$

$\Rightarrow$ $\boxed{c = 0}$

$\therefore$ Required soln is

$F(\alpha) = \pi\alpha$

$\Rightarrow$ $\int_0^{\pi} \frac{\log(1+\sin\alpha \cdot \cos x)}{\cos x}\, dx = \pi\alpha$.

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Ex: prove that $\int_0^\infty \frac{\tan^{-1} ax}{x(1+x^2)}\,dx = \pi/2 \log(1+a)$

By Differentiating under the integral.

Soln : let $I = \int_0^\infty \frac{\tan^{-1} ax}{x(1+x^2)} \cdot dx = F(a)$ —(1)

where x is variable & a is parameter

∴ Diff. w.r.to a on both side we get

$$\frac{dF(a)}{da} = \frac{d}{da}\left[\int_0^\infty \frac{\tan^{-1} ax}{x(1+x^2)}\,dx\right]$$

Since limit of integration are not depend on parameter a

∴ By Leibnitz rule (1) of Diff under integral, we get

$$\frac{dF(a)}{da} = \int_{x=0}^\infty \left[\frac{\partial}{\partial a}\left(\frac{\tan^{-1} ax}{x(1+x^2)}\right)\right]dx$$

$$= \int_{x=0}^\infty \frac{1}{x(1+x^2)} \cdot \left[\frac{\partial}{\partial a}(\tan^{-1} ax)\right] dx$$

$$= \int_{x=a}^\infty \frac{1}{x(1+x^2)} \cdot \left[\frac{1}{1+a^2x^2}(x)\right] dx$$

$$= \int_{x=a}^\infty \frac{1}{(1+x^2)\cdot(1+a^2x^2)} \cdot dx$$

$$= \frac{1}{(1-a^2)} \int_0^\infty \left[\frac{1}{(1+x^2)} - \frac{a^2}{(1+a^2x^2/a^2)}\right] dx$$ (By partial fraction method)

$$= \frac{1}{(1-a^2)} \left\{\left[\tan^{-1} x\right]_0^\infty - \left[\frac{1}{(1/a)} \cdot \tan^{-1}\left(\frac{x}{1/a}\right)\right]_0^\infty\right\}$$

$$= \frac{1}{(1-a^2)} \left\{\left[\tan^{-1}\infty - \tan^{-1} 0\right] - \left[a\tan^{-1}(\infty) - \tan^{-1} 0\right]\right\}$$

$$= \frac{1}{(1-a^2)} \left[\pi/2 - a\pi/2\right] = \frac{(1-a)}{(1-a^2)}\pi/2 = \frac{1}{(1+a)}\frac{\pi}{2}$$

$$\boxed{\frac{dF(a)}{da} = \frac{\pi}{2(1+a)}}$$

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Now separating variables

$$dF(a) = \frac{\pi}{2(1+a)}\,da$$

on integration both side

$$\int dF(a) = \frac{\pi}{2}\int \frac{1}{(1+a)}\,da + c$$

$$\Rightarrow F(a) = \pi/2 \log(1+a) + c$$

$\therefore$ Using (1)

$$\Rightarrow \boxed{\int_0^\infty \frac{\tan^{-1} ax}{x(1+x^2)}\,dx = \pi/2 \log(1+a) + c} \quad \text{---(2)}$$

To find c

put $a=0$ in above eqn, we get

$$\int_0^\infty \frac{\tan^{-1} 0}{x(1+x^2)}\,dx = \pi/2 \log(1) + c$$

$$\Rightarrow \quad 0 = \pi/2\,(0) + c \qquad \Rightarrow \boxed{c=0}$$

$\therefore$ Req^d value of given integral is

$$\boxed{\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx = \pi/2 \log(1+a)}$$

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H.W

H.W :

① Evaluate $F(\alpha) = \int_0^{\pi} \frac{\log(1+\alpha\cos x)}{\cos x}\, dx$

where $|\alpha| < 1$ Ans $F(\alpha) = \pi \sin^{-1}\alpha$

Imp ② prove that $\int_0^{\infty} \frac{\tan^{-1} ax}{x(1+x^2)}\, dx = \frac{\pi}{2}\log(1+a)$, $a > 0$

by differentiating under integral sign.

③ Evaluate $\int_0^1 \frac{x^a - x^b}{\log x}\, dx$ $a > 0$, $b > 0$

Hint: $\int_0^1 \frac{x^a - x^b}{\log x}\, dx = \left[\int_0^1 \frac{x^a}{\log x}\, dx\right] - \left[\int_0^1 \frac{x^b}{\log x}\, dx\right]$

$= F(a) - F(b)$

Solve for $F(a)$

Similarly for $F(b)$

$= \log(a+1) - \log(b+1)$

Ans $\log\left(\frac{a+1}{b+1}\right)$

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Problem based on Leibnitz rule (II)

Ex Using ~~for~~ leibnitz rule for differentiation under integral sign. ~~evaluate~~ Find value of $\frac{dF(\alpha)}{d\alpha}$.

if $F(\alpha) = \int_0^{\alpha^2} \tan^{-1}\left(\frac{x}{\alpha}\right) dx.$

also verify:

Soln :⇒ let $F(\alpha) = \int_0^{\alpha^2} \tan^{-1}\left(\frac{x}{\alpha}\right) dx$ ——— (1)

Diff. w.r.to. α we get

$$\frac{dF(\alpha)}{d\alpha} = \frac{d}{d\alpha}\left[\int_0^{\alpha^2} \tan^{-1}\left(\frac{x}{\alpha}\right) dx\right]$$

Using leibnitz-rule -II

$$= \int_0^{\alpha^2} \frac{\partial}{\partial\alpha}\left(\tan^{-1}\left(\frac{x}{\alpha}\right)\right) dx + \frac{d}{d\alpha}(\alpha^2) \cdot \tan^{-1}\left(\frac{\alpha^2}{\alpha}\right) - \frac{d}{d\alpha}(0) \cdot \tan^{-1}\left(\frac{0}{\alpha}\right)$$

$$= \int_0^{\alpha^2} \frac{1}{1+\left(\frac{x}{\alpha}\right)^2}\left(\frac{-x}{\alpha^2}\right) dx + 2\alpha \cdot \tan^{-1}(\alpha) - 0$$

$$= \int_0^{\alpha^2} \frac{-x}{\alpha^2+x^2} dx + 2\alpha \tan^{-1}\alpha$$

$$=$$ put $x^2 = t \Rightarrow 2x\,dx = dt$

as $x=0 \Rightarrow t=0$

$x=\alpha^2 \Rightarrow t=\alpha^4$

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$$\therefore \frac{dF(\alpha)}{d\alpha} = -\int_0^{\alpha^4} \frac{1}{\alpha^2+t} \cdot \frac{dt}{2} + 2\alpha \tan^{-1}\alpha$$

$$= -\frac{1}{2}\left[\log(\alpha^2+t)\right]_0^{\alpha^4} + 2\alpha\tan^{-1}\alpha$$

$$= -\frac{1}{2}\left[\log(\alpha^2+\alpha^4) - \log(\alpha^2)\right] + 2\alpha\tan^{-1}\alpha$$

$$= \frac{1}{2}\log\left(\frac{\alpha^2+\alpha^4}{\alpha^2}\right) + 2\alpha\tan^{-1}\alpha$$

$$\boxed{\frac{dF(\alpha)}{d\alpha} = \frac{1}{2}\log(1+\alpha^2) + 2\alpha\tan^{-1}\alpha}$$

$$= +\log(1+\alpha^2)^{-1/2} + 2\alpha\tan^{-1}\alpha$$

$$\boxed{\frac{dF(\alpha)}{d\alpha} = \log\left(\frac{1}{\sqrt{1+\alpha^2}}\right) + 2\alpha\tan^{-1}\alpha}$$ Ans

Verification

$$\frac{dF(\alpha)}{d\alpha} = \frac{d}{d\alpha}\left[\int_0^{\alpha^2} \tan^{-1}(x/\alpha)\, dx\right]$$

$$= \frac{d}{d\alpha}\cdot\left[\int_0^{\alpha^2} \tan^{-1}(x/\alpha)\cdot 1\, dx\right]$$

LIATE

I · II

integrating by parts we get

$$= \frac{d}{d\alpha}\left\{\left[\tan^{-1}(x/\alpha)\cdot\int 1\, dx\right]_0^{\alpha^2} - \int_0^{\alpha^2}\left[\frac{d}{dx}(\tan^{-1}(x/\alpha))\cdot\int 1\, dx\right]dx\right\}$$

$$= \frac{d}{d\alpha}\left\{\left[\tan^{-1}\left(\frac{x}{\alpha}\right)\cdot x\right]_0^{\alpha^2} - \int_0^{\alpha^2}\frac{1}{1+(x/\alpha)^2}\cdot\left(\frac{1}{\alpha}\right)\cdot x\, dx\right\}$$

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$$= \frac{d}{d\alpha}\left[\left(\alpha^2 \tan^{-1}(\alpha) - 0\right) - \int_0^{\alpha^2} \frac{\alpha x}{x^2+\alpha^2}\, dx\right]$$

$$= \frac{d}{d\alpha}\left[\alpha^2 \tan^{-1}(\alpha) - \frac{\alpha}{2} \cdot \log(x^2+\alpha^2)\Big|_0^{\alpha^2}\right]$$

$$= \frac{d}{d\alpha}\left[\alpha^2 \tan^{-1}\alpha - \alpha/2 \left(\log(\alpha^2+\alpha^2) - \log(\alpha^2)\right)\right]$$

$$= \frac{d}{d\alpha}\left[\alpha^2 \tan^{-1}\alpha - \frac{\alpha}{2} \log(1+\alpha^2)\right]$$

$$= \alpha^2 \frac{1}{1+\alpha^2} + 2\alpha \cdot \tan^{-1}\alpha - \frac{\alpha}{2} \cdot \frac{1}{1+\alpha^2}(2\alpha) - \frac{1}{2}\log(1+\alpha^2)$$

$$= 2\alpha \tan^{-1}\alpha - \frac{1}{2}\log(1+\alpha^2)$$

$$\boxed{\frac{df(\alpha)}{d\alpha} = 2\alpha \tan^{-1}\alpha + \log\frac{1}{\sqrt{1+\alpha^2}}}$$

Hence verified.

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Ex: Evaluate $\int_0^{\alpha} \frac{\log(1+\alpha x)}{1+x^2}\,dx$

Hence show that $\int_0^{1} \frac{\log(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\log_e 2$

Soln: let

$$F(\alpha) = \int_0^{\alpha} \frac{\log(1+\alpha x)}{(1+x^2)}\,dx = \int_{\psi(\alpha)}^{\phi(\alpha)} f(x,\alpha)\,dx \quad \text{---(1)}$$

where $f(x,\alpha) = \frac{\log(1+\alpha x)}{(1+x^2)}$, $\phi(x)=\alpha$, and $\psi(\alpha)=0$

$$\Rightarrow \frac{d\phi(\alpha)}{d\alpha} = 1, \quad \frac{d\psi(\alpha)}{d\alpha} = 0$$

$$f(\phi(\alpha),\alpha) = \log f(\alpha,\alpha) = \frac{\log(1+\alpha^2)}{(1+\alpha^2)}$$

$$f(\psi(\alpha),\alpha) = f(0,\alpha) = \frac{\log(1+0)}{(1+0)} = 0$$

$\therefore$ Now. Diff. under integral sign., we get

$$\frac{dF(\alpha)}{d\alpha} = \frac{d}{d\alpha}\int_0^{\alpha} \frac{\log(1+\alpha x)}{(1+x^2)}\,dx$$

$\therefore$ By Leibnitz rule-II. (as limit of integral is function of parameter α.)

$$\therefore \frac{dF(\alpha)}{d\alpha} = \int_0^{\alpha} \frac{\partial}{\partial\alpha}\left(\frac{\log(1+\alpha x)}{(1+x^2)}\right)dx + \frac{d}{d\alpha}(\alpha)\cdot f(\alpha,\alpha) - \frac{d}{d\alpha}(0)\cdot f(0,\alpha)$$

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$$= \left[\int_0^{\alpha} \frac{1}{(1+x^2)} \cdot \frac{1}{(1+\alpha x)} \cdot x \, dx \right] + \frac{1}{(1+\alpha^2)} \log(1+\alpha^2) - 0$$

$$\frac{dF(\alpha)}{d\alpha} = \frac{\log(1+\alpha^2)}{(1+\alpha^2)} + \int_0^{\alpha} \frac{x}{(1+x^2)(1+\alpha x)} \, dx \quad \text{---(2)}$$

By partial fraction method.

$$\frac{x}{(1+x^2)(1+\alpha x)} = \frac{Ax+B}{(1+x^2)} + \frac{C}{(1+\alpha x)}$$

$$\Rightarrow \quad x = (Ax+B)(1+\alpha x) + C(1+x^2)$$

$$\Rightarrow \quad x = \alpha A x^2 + (A+\alpha B)x + B + C + Cx^2$$

$$x = (\alpha A + C)x^2 + (A+\alpha B)x + (B+C)$$

equating ~~like power of~~ coefficient of like power of x

$$\therefore \quad \alpha A + C = 0 \quad \Rightarrow \quad C = -\alpha A \quad \text{---(1)}$$

$$A + \alpha B = 1 \quad \text{---(2)}$$

$$B + C = 0 \quad \text{---(3)}$$

using (1) & (3) $\Rightarrow$ $B - \alpha A = 0$ ---(4)

Solving (2) & (4) we get [eqn(2) $\times \alpha$ + eqn(4)]

$$2A + \alpha^2 B = \alpha$$

$$+ \quad B - \alpha A = 0$$

$$B(\alpha^2+1) = \alpha$$

$$\therefore \boxed{B = \frac{\alpha}{\alpha^2+1}}$$

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$$\therefore \boxed{C = \frac{-\alpha}{\alpha^2+1}}$$

from (2)

$$A = 1 - \alpha B = 1 - \frac{\alpha^2}{(1+\alpha^2)}$$

$$\boxed{A = \frac{1}{1+\alpha^2}}$$

$$\therefore (2) \Rightarrow \frac{dF(\alpha)}{d\alpha} = \frac{\log(1+\alpha^2)}{(1+\alpha^2)} + \int_0^\alpha \left[\frac{1}{(1+\alpha^2)} \frac{x}{(1+x^2)} + \frac{\alpha}{(1+\alpha^2)} \cdot \frac{1}{(1+x^2)} - \frac{\alpha}{\alpha^2+1} \cdot \frac{1}{1+\alpha x} \right] dx$$

$$= \frac{\log(1+\alpha^2)}{1+\alpha^2} + \frac{1}{(1+\alpha^2)} \left(\int_0^\alpha \frac{x}{(1+x^2)} dx \right)$$

$$+ \frac{\alpha}{1+\alpha^2} \left(\int_0^\alpha \frac{1}{1+x^2} dx \right) - \frac{\alpha}{\alpha^2+1} \left(\int_0^\alpha \frac{1}{1+\alpha x} dx \right)$$

$$= \frac{\log(1+\alpha^2)}{(1+\alpha^2)} + \frac{1}{(1+\alpha^2)} \frac{1}{2} \left[\log(1+x^2) \right]_0^\alpha$$

$$+ \frac{\alpha}{1+\alpha^2} \left[\tan^{-1} x \right]_0^\alpha - \frac{\alpha}{\alpha^2+1} \left[\log(1+\alpha x) \right]_0^\alpha$$

$$= \frac{\log(1+\alpha^2)}{(1+\alpha^2)} + \frac{1}{2} \frac{\log(1+\alpha^2)}{(1+\alpha^2)} + \frac{\alpha \tan^{-1}\alpha}{1+\alpha^2} - \frac{\log(1+\alpha^2)}{(1+\alpha^2)}$$

$$\frac{dF(\alpha)}{d\alpha} = \frac{1}{2(1+\alpha^2)} \cdot \left[\frac{1}{2} \log(1+\alpha^2) + 2\alpha \tan^{-1}\alpha \right]$$

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$$dF(\alpha) = \frac{1}{1+\alpha^2}\left[\frac{1}{2}\log(1+\alpha^2) + \alpha\tan^{-1}\alpha\right] d\alpha$$

on integrating ~~w.r.t.~~ w.r.to. α, we get

$$F(\alpha) = \frac{1}{2}\int \underset{I}{\log(1+\alpha^2)} \cdot \underset{II}{\frac{1}{(1+\alpha^2)}}\, d\alpha$$

$$+ \int \tan^{-1}\alpha \cdot \frac{\alpha}{(1+\alpha^2)}\, d\alpha + C$$

$$= \frac{1}{2}\left\{\left[\log(1+\alpha^2)\cdot\tan^{-1}\alpha\right] - \int \frac{2\alpha}{(1+\alpha^2)}\cdot\tan^{-1}\alpha\, d\alpha\right\}$$

$$+ \int \tan^{-1}\alpha \cdot \frac{\alpha}{1+\alpha^2}\, d\alpha + C$$

$$= \frac{1}{2}\log(1+\alpha^2)\cdot\tan^{-1}\alpha - \int \frac{\alpha}{(1+\alpha^2)}\tan^{-1}\alpha\, d\alpha$$

$$+ \int \tan^{-1}\alpha \cdot \frac{\alpha}{1+\alpha^2}\, d\alpha + C$$

$$\boxed{F(\alpha) = \frac{1}{2}\log(1+\alpha^2)\tan^{-1}\alpha + C} \quad \text{---------- (3)}$$

To find const. C

put $\alpha = 0$ in eqn (1) & (3)

(1) $\Rightarrow$ $F(0) = 0$

(3) $\Rightarrow$ $F(0) = \frac{1}{2}\log(1+0)\tan^{-1}0 + C \Rightarrow F(0) = C$

$\therefore \Rightarrow \boxed{C = 0}$

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Hence Reqd soln is

$$F(\alpha) = \frac{1}{2}\log(1+\alpha^2)\cdot\tan^{-1}\alpha$$

i.e. $$\int_0^{\alpha} \frac{\log(1+x)}{(1+x^2)}\,dx = \frac{1}{2}\log(1+\alpha^2)\cdot\tan^{-1}\alpha.$$

Now,

Put $\alpha = 1$

$$\therefore \int_0^1 \frac{\log(1+x)}{(1+x^2)}\,dx = \frac{1}{2}\log(1+1)\cdot\tan^{-1}1$$

$$= \frac{1}{2}\cdot\frac{\pi}{4}\cdot\log(2) = \frac{\pi}{8}\log(2).$$

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Ex: Using differentiation under integral sign.

Show that $\int_{\pi/6a}^{\pi/2a} \frac{\sin ax}{x}\,dx$ is independent of a,

Soln: let

$$F(a) = \int_{\pi/6a}^{\pi/2a} \frac{\sin ax}{x}\,dx \quad \text{---} (1)$$

Differentiating w.r.to a. we get

$$\frac{dF(a)}{da} = \frac{d}{da} \int_{\pi/6a}^{\pi/2a} \frac{\sin ax}{x}\,dx$$

$\therefore$ By leibnitz rule - II ($\because$ Cos limits are function of a).

$$\frac{dF(a)}{da} = \int_{\pi/6a}^{\pi/2a} \frac{\partial}{\partial a}\left(\frac{\sin ax}{x}\right)dx + \frac{d}{da}(\pi/2a) \cdot \frac{\sin(a \cdot \pi/2a)}{(\pi/2a)}$$

$$- \frac{d}{da}\left(\frac{\pi}{6a}\right) \cdot \frac{\sin\left(a \cdot \frac{\pi}{6a}\right)}{\left(\frac{\pi}{6a}\right)}$$

$$= \int_{\pi/6a}^{\pi/2a} \frac{1}{x} \cdot \cos ax \cdot x\,dx + \frac{-\pi}{2a^2} \cdot \frac{2a}{\pi} \sin(\pi/2)$$

$$+ \frac{\pi}{6a^2} \cdot \frac{6a}{\pi} \cdot \sin\left(\frac{\pi}{6}\right)$$

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$$= \int_{\pi/6a}^{\pi/2a} \cos ax \cdot dx + \left(\frac{-1}{a}\right)(1) + \left(\frac{1}{a}\right)\left(\frac{1}{2}\right)$$

$$= \left[\frac{\sin ax}{a}\right]_{\pi/6a}^{\pi/2a} + \left(\frac{-1}{2a}\right)$$

$$= \frac{1}{a}\left[\sin(\pi/2) - \sin \pi/6\right] + \frac{-1}{2a}$$

$$= \frac{1}{a}\left[1 - \frac{1}{2}\right] + \left(\frac{-1}{2a}\right)$$

$$= \frac{1}{2a} - \frac{1}{2a} = 0$$

$$\Rightarrow \boxed{\frac{dF(a)}{da} = 0}$$

$\Rightarrow$ F(a) is not depend on a.

ie. $\int_{\pi/6a}^{\pi/2a} \left(\frac{\sin ax}{x}\right) dx$ is indpendent on a.

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Find $\frac{d^2F(\alpha)}{d\alpha^2}$ if $F(\alpha) = \int_{1/\alpha}^{2/\alpha} \frac{\sin \alpha x}{x^2}\, dx$

using differentiation under integral sign

$\left| \frac{d^2F(\alpha)}{d\alpha} = \frac{d}{d\alpha}\left(\frac{dF}{d\alpha}\right) \right.$

Soln: Given

$$F(\alpha) = \int_{1/\alpha}^{2/\alpha} \frac{\sin \alpha x}{x^2}\, dx \quad \longrightarrow (1)$$

Diff. w.r.to. α we get-

$$\frac{dF(\alpha)}{d\alpha} = \frac{d}{d\alpha}\left[\int_{1/\alpha}^{2/\alpha} \frac{\sin \alpha x}{x^2}\right] dx.$$

$\therefore$ Diff. under integral sign. (using Leibnitz rule - II)

$$\frac{dF(\alpha)}{d\alpha} = \int_{1/\alpha}^{2/\alpha} \frac{\partial}{\partial \alpha}\left(\frac{\sin \alpha x}{x^2}\right) \cdot dx + \frac{d}{d\alpha}\left(\frac{2}{\alpha}\right) \cdot \frac{\sin(x \cdot 2/\alpha)}{(2/\alpha)^2}$$

$$- \frac{d}{d\alpha}\left(\frac{1}{\alpha}\right) \cdot \frac{\sin(\alpha \cdot 1/\alpha)}{(1/\alpha)^2}$$

$$\frac{dF(\alpha)}{d\alpha} = \int_{1/\alpha}^{2/\alpha} \left(\frac{1}{x^2}\cos \alpha x - x\right) dx + \left(\frac{-2}{\alpha^2}\right)(\sin 2) \cdot \frac{\alpha^2}{4}$$

$$- \left(\frac{-1}{\alpha^2}\right) \sin(1) \cdot (\alpha^2)$$

$$\frac{dF(\alpha)}{d\alpha} = \int_{1/\alpha}^{2/\alpha} \frac{\cos \alpha x}{x}\, dx + \frac{-1}{2}\sin(2) + \sin 1.$$

again diff. w.r.t α we get

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$$\frac{d}{d\alpha}\left(\frac{dF(\alpha)}{d\alpha}\right) = \frac{d}{d\alpha}\left[\int_{1/\alpha}^{2/\alpha} \frac{\cos\alpha x}{x}\,dx + \frac{1}{2}\sin(2) + \sin(1)\right]$$

$$\frac{d^2F(\alpha)}{d\alpha^2} = \frac{d}{d\alpha}\left[\int_{1/\alpha}^{2/\alpha} \frac{\cos\alpha x}{x}\,dx\right] - 0 + 0$$

again diff. under integral sign (using leibnitz rule II)

$$\frac{d^2F(\alpha)}{d\alpha^2} = \int_{1/\alpha}^{2/\alpha} \left[\frac{\partial}{\partial\alpha}\left(\frac{\cos\alpha x}{x}\right)\right]dx + \frac{d}{d\alpha}\left(\frac{2}{\alpha}\right)\cdot\cos\left(\frac{\alpha\cdot 2\alpha}{2/\alpha}\right)$$

$$- \frac{d}{d\alpha}\left(\frac{1}{\alpha}\right)\cos\left(\frac{\alpha\cdot 1/\alpha}{1/\alpha}\right)$$

$$= \int_{1/\alpha}^{2/\alpha} \frac{1}{x}(-\sin\alpha x)\cdot x\,dx - \frac{2}{\alpha^2}\left(\frac{\alpha}{2}\right)\cos(2)$$

$$+ \frac{1}{\alpha^2}(\alpha)\cos(1)$$

$$= -\int_{1/\alpha}^{2/\alpha} \sin\alpha x\,dx - \frac{1}{\alpha}\cos 2 + \frac{1}{\alpha}\cos 1$$

$$= -\left[\frac{-\cos\alpha x}{\alpha}\right]_{1/\alpha}^{2/\alpha} - \frac{1}{\alpha}\cos 2 + \frac{1}{\alpha}\cos 1$$

$$= +\left[\frac{\cos 2(2/\alpha)}{\alpha} - \frac{\cos(\alpha\cdot 1/\alpha)}{\alpha}\right] - \frac{1}{\alpha}\cos(2) + \frac{1}{\alpha}(\cos 1)$$

$$= \frac{1}{\alpha}\cos 2 - \frac{1}{\alpha}\cos(1) - \frac{1}{\alpha}\cos(2) + \frac{1}{\alpha}\cos(1)$$

$$= 0$$

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Curve Tracing :⇒

Curve tracing is a procedure to obtain an approximate shape of the curve without plotting a large number of points on it.

* Tracing of cartesian curve $f(x,y)=0$

The following point are to be taken into consideration while tracing a cartesian curve.

(i) Symmetry :⇒

(a) A curve $f(x,y)=0$ is symmetric about x-axis (line $y=0$) if curve contain terms of even power of y only.

i.e $f(x,y) = f(x,-y)$

for eg: $y^2(a-x^3) = y^4+x^2$

(b) A curve $f(x,y)=0$ is symmetric about y-axis (line $x=0$) if curve contain terms of even power of x only.

i.e. $f(x,y) = f(-x,y)$

for eg: (i) $x^2 = 4ay$

(ii) $y^2(a+x^2) = x^2(a-x^2)$ etc.

(c) A curve is symmetric both co-ordinator axis i.e. x-axis and y axis if it contain only even power of x & y both

i.e $f(x,y) = f(-x,-y)$

for eg: (i) $y^2(a+x^2) = x^2(a-x^2)$ (ii) $x^2+y^2=a^2$

(iii)

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(d) Curve is symmetric to the line $y=x$ if the curve remain unchanged by interchanging x and y.

i.e $f(x,y) = f(y,x)$.

For eg: (i) $x^3+y^3 = 3axy$

(ii) $x^2+y^2 = 2$

(e) A curve is symmetric in opposite quadrant if curve remain unaltered after replacing x by $-x$ and y by $-y$. and curve does not contain even powers of x and y only.

foreg: $x^5+y^5 = 5x^2y$.

(2) Origin: [Nature of origin.]

A curve is passing through an origin $(0,0)$ iff it does not contain any constant term

foreg: (i) $x^3+y^3 = 3axy$

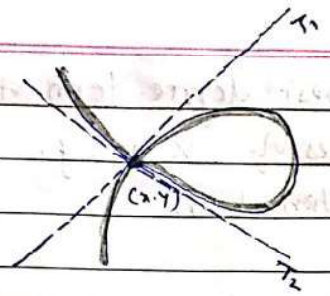
(ii) $y^2(a-x) = x^3$

★ Tangent at origin:

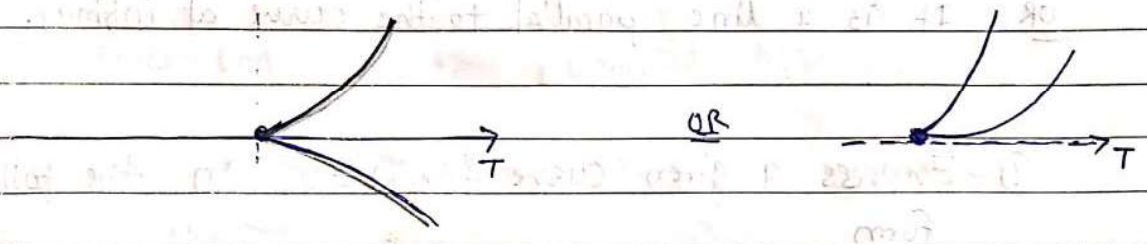
Tangent at origin can be obtained by equating lowest degree term of given curve to zero.

(i) if at Point (x,y), two real and distinct tangents are obtained then pt (x,y) is called a node and it is represented as follows.

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(ii) If at pt (x,y), a tangent is real and coinsident (ie same the point is called as **Cusp**.



(3) **Point of Intersection :⇒**

The point of intersection (x,y) can be obtained as follows

(i) If curve is symmetric about x-axis. (ie y=0)
put y=0 in given curve and find x.

(ii) If curve is symmetric about y-axis (i.e. x=0)
put x=0 in given curve and find y.

(iii) If curve is symmetric to line y=x
put x=y in given curve and find x. and Hence y.

Tangent at point of intersection say (h,k)

Replace x by X+h and y by Y+k in the given equation of curve. giving

$$f(X+h, Y+k) = 0$$

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the equate lowest degree term to zero, Hence find values of values of x and y. giving a tangent at point of intersection (h,k).

(4) Asymtotes :⇒

It is a tangent at infinity to the curve.

OR It is a line parallel to the curve at infinity.

(i) Express a given curve $f(x,y)=0$ in the following form

$$y=f(x) \quad [\text{OR } x=f(y)]$$

If for a particular value of $x=a$ say, y tends to infinite, Then $x=a$ is called as asymtotes to the given curve

for eg: $yx-ya = x^2 \Rightarrow y(x-a)=x^2$

$$\Rightarrow y = \frac{x^2}{(x-a)}$$

$\because$ at $x=a \Rightarrow y\to\infty$

$\therefore$ $x=a$ is an asymtotes.

(ii) If given curve $f(x,y)$ is implicit. (i.e. it is not possible to represent it in any of the following form. $y=f(x)$ or $x=f(y)$.)

then line $y=mx+c$ is an asymtotes to the given curve, which is obtained as explain below.

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let $\phi_n(x,y)$ = highest degree term in the curve.

$\phi_{n-1}(x,y)$ = second highest degree term in the curve

Then replace x by 1 and y by m we get

$$\phi_n(1,m) = \phi_n(m) \quad , \quad \phi_{n-1}(1,m) = \phi_{n-1}(m)$$

Hence find $\phi_n'(m)$ provided $\phi_n'(m) \neq 0$.

(if $\phi_n'(m) = 0$ then there is no asymtotes.)

Hence find value of $\boxed{c = -\frac{\phi_{n-1}(m)}{\phi_n'(m)}}$

Now Consider $\phi_n(m) = 0$ (polynominal in m)
and find its roots i.e. $m_1, m_2, m_3 \ldots$
also find corresponding values of c's. s.t. $c_1, c_2, c_3 \ldots$ giving an asymtotes
$y = m_1x + c_1$, $y = m_2x + c_2$ ... and soon.

(5) Region :⇒ (If curve is explicit)
let $f(x,y) = 0$ be a given curve as express as follows. $y = f(x)$. [or $x = g(y)$]

Then value of $x = \alpha$ (say) for which $y = f(x)$ become imaginary represent that given cosve does not exist in beyond the line $x = \alpha$.

OR

(If given curve is implicit. (or Explicit.), the region of existance is lies betn tangent at point of intersection and asymtotes to the curve)

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Ex: Trace a curve $y^2(a+x) = x^2(a-x)$

Soln :⇒ Given curve is

$$y^2(a+x) = x^2(a-x) \quad \text{---(1)}$$

$$\Rightarrow y = \sqrt{\frac{x^2(a-x)}{(a+x)}}$$

(i) **symmetry :**

Since eqn (1) contain only even power of y.

⇒ It is symmetric to the line $y=0$ i.e. x-axis.

(ii) **origin :** Since eqn (1) does not contain any const. term

⇒ it passing through the origin $(0,0)$.

Tangent at origin :

given curve is $y^2(a+x) = x^2(a-x)$

$$\Rightarrow y^2a + y^2x = x^2a - x^3$$

$$\Rightarrow y^2x + x^3 + a(y^2 - x^2) = 0$$

equating lowest degree term to zero

i.e. $a(y^2 - x^2) = 0$

$$\Rightarrow y^2 = x^2$$

$$\boxed{y = \pm x}$$

Thus tangent at origin is $y = +x$ and $y = -x$ are real and distinct. and Hence origin is a node

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(iii) Point of intersection

Since given curve is symmetric to line $y=0$ (i.e. x-axis)

$\therefore$ we put $y=0$ in eqn (1), we get

$$x=a$$

$\therefore$ point of intersection $(h,k) = (a,0)$

Tangent at origin:

Put $x = X+a, \quad y = Y+0$

$\therefore$ (1) $\Rightarrow$ $Y^2[a+(X+a)] = (X+a)^2[a-(X+a)]$

$\Rightarrow$ $X^3 + 2aXY^2 + 2aY^2 + 2aX^2 + a^2X = 0$

equating lowest degree term to zero., we get

$$a^2x = 0 \Rightarrow x = 0$$

i.e. $x-a=0$

$\Rightarrow x=a$

Thus line $x=a$ is a tangent at point $(a,0)$.

(iv) Asymtotes:

Since $y = \sqrt{\frac{x^2(a-x)}{(a+x)}}$ tends to ∞

if we put $x=-a$

$\therefore$ line $x=-a$ is an asymtoes to the curve.

(v) Region: since

$$y = \sqrt{\frac{x^2(a-x)}{(a+x)}}$$

is become an imaginary when $x>a$ and $x<-a$

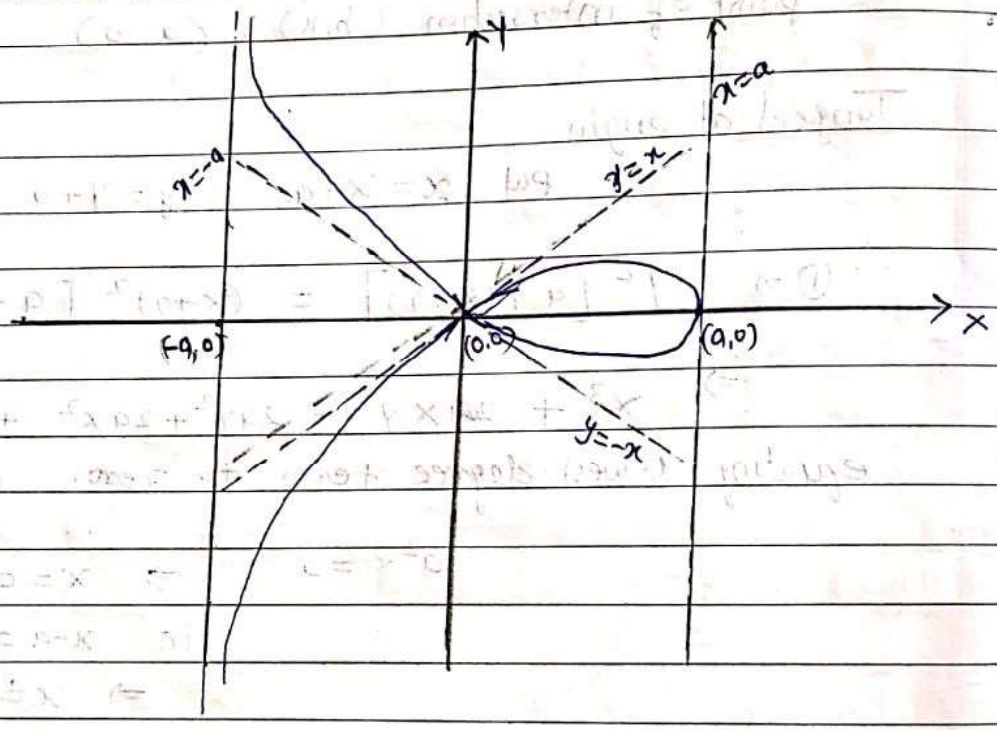
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Thus curve must be exist in the region

$-a < x < a$

Curve Representation



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Ex: Trace a curve $x^3+y^3 = 3axy$.

Soln: Given eqn of curve

$$x^3+y^3 = 3axy \quad \text{---(1)}$$

(i) Symmetry:

since if we put $x=y$. the eqn of curve remain unaltered

$\Rightarrow$ it is symmetrical to the line $y=x$

(ii) origin: since curve does not contain any constant term

$\Rightarrow$ it passing through the origin.

Tangent at origin:

equating lowest degree term to zero.

i.e. $3axy = 0$

$\Rightarrow x=0$ & $y=0$

i.e. $x=0$ (y-axis) and $y=0$ (x-axis) are two real and distinct tangents of the curve at origin.

$\therefore$ origin is a node

(iii) point of intersection:

Since curve is symmetric to line $y=x$

$\therefore$ put $y=x$ in given eqn of curve.

we get $y^3+y^3 = 3ay^2$

$\Rightarrow y = \frac{3a}{2}$ Hence $x = \frac{3a}{2}$

$\therefore$ point of intersection $(h,k) = (3a/2, 3a/2)$.

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Tangent at point of intersection $(3a/2, 3a/2)$

put $x = X + 3a/2$ & $y = Y + 3a/2$ in eq^n (1)

$\therefore$ (1) $\Rightarrow$ $\left(X+\frac{3a}{2}\right)^3 + \left(Y+\frac{3a}{2}\right)^3 = 3a\left(X+\frac{3a}{2}\right)\left(Y+\frac{3a}{2}\right)$

$\Rightarrow \left(X^3 + \frac{9}{2}X^2a + \frac{27}{4}a^2X + \frac{27}{8}a^3\right) + \left(Y^3 + \frac{9a}{2}Y^2 + \frac{27}{4}a^2Y + \frac{27}{8}a^3\right)$

$= 3aXY + \frac{9a^2}{2}X + \frac{9a^2}{2}Y + \frac{27a^3}{4}$

$\Rightarrow (X^3+Y^3) + \frac{9a}{2}X^2 + \frac{9a}{2}Y^2 + 3aXY + \frac{27}{4}a^2(X+Y) + \frac{9a^2}{2}(X+Y) = 0$

equating lowest degree term to zero

i.e. $\frac{27}{4}a^2(X+Y) - \frac{9a^2}{2}(X+Y) = 0$

$\Rightarrow \left(\frac{27}{4} - \frac{9}{2}\right)a^2(X+Y) = 0$

$\Rightarrow X+Y = 0$

$\Rightarrow \left(x - \frac{3a}{2}\right) + \left(y - \frac{3a}{2}\right) = 0$

$\Rightarrow x + y - 3a = 0$

$\Rightarrow x + y = 3a$

Thus tangent at pt $(3a/2, 3a/2)$ is $x+y=3a$

x	0	3a
y	3a	0

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(iv) **Asymtotes** :⇒ Since given curve can not be express interms of x or interms of y. (i.e. Implicit Form).
there is no asymtotes parallel to x and y axis.
∴ it may have oblique asymtotes. $\boxed{y = mx + c}$

let $\phi_n(x,y) = x^3 + y^3 \Rightarrow \phi_n(1,m) = 1 + m^3 = \phi_n(m)$

$\phi_{n-1}(x,y) = -3axy \Rightarrow \phi_{n-1}(1,m) = -3am = \phi_{n-1}(m)$

also $\phi_n'(m) = 3m^2$

$\therefore c = -\dfrac{\phi_{n-1}(m)}{\phi_n'(m)} = \dfrac{-(-3am)}{3m^2} = \dfrac{a}{m}$

Now put

$\phi_n(1,m) = 0 \Rightarrow 1 + m^3 = 0$

$\Rightarrow m^3 = -1 \Rightarrow \boxed{m = -1}$

thus $\boxed{c = -a}$

Hence req^d asymtotes is $y = mx + c = -x - a$

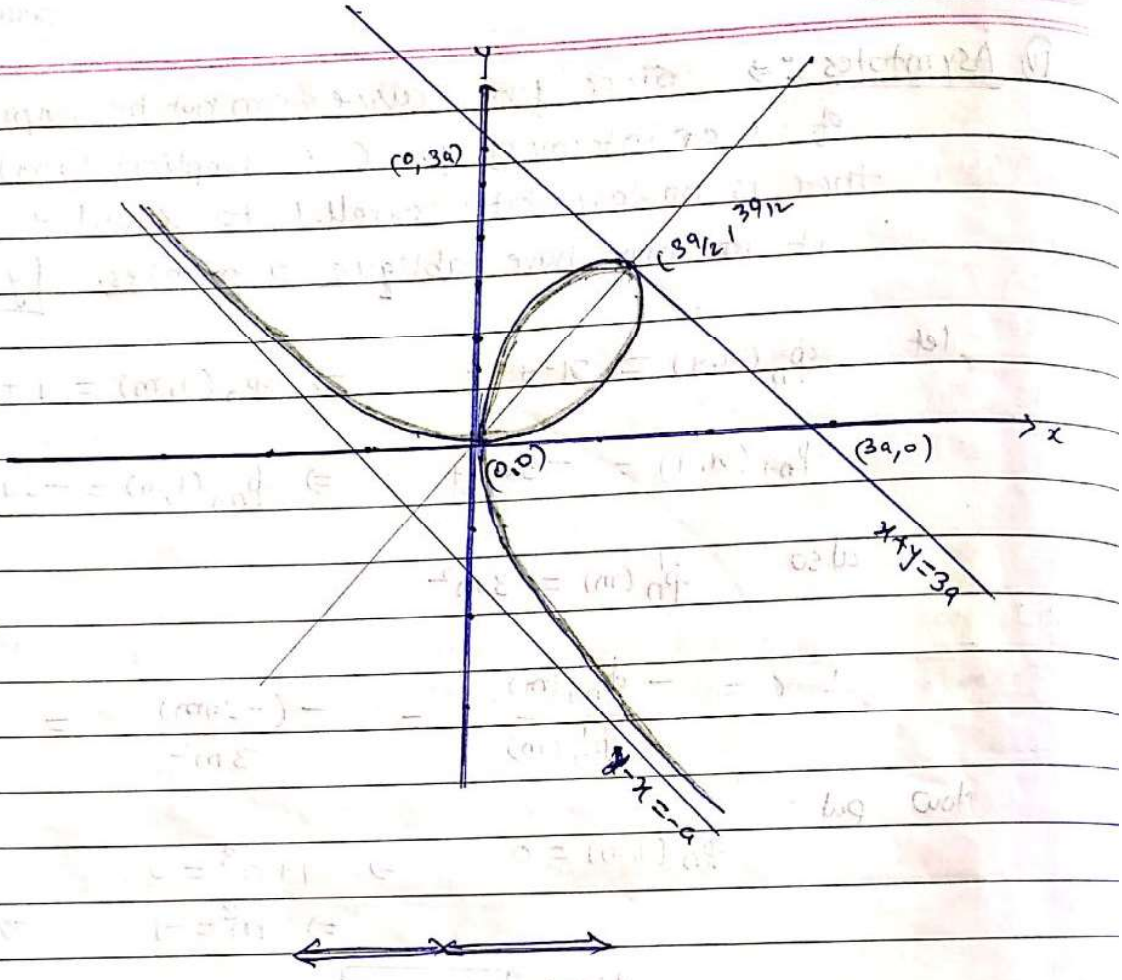
$\Rightarrow y + x = -a.$

x	0	$-a$
y	$-a$	0

(v) **Region**: Since given curve is implicit.
Region of curve is lies betn tangent at point of intersection and asymtotes of the curve.

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Ex: Trace a curve $y^2(4-x) = x^3$

Soln:

Given curve is $y^2(4-x) = x^3$ ——— (1)

$$\Rightarrow y = \sqrt{\frac{x^3}{(4-x)}}$$

(i) **Symmetry:**

Since curve have even power of y only.

$\Rightarrow$ it is symmetrical to line $y=0$ (i.e. x-axis)

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(ii) Origin:

Since curve does not contain any const. term.
⇒ it is passing through the origin (0,0)

Tangent at origin:

equating lowest degree term to zero

$4y^2 = 0 \Rightarrow y = 0$ & $y = 0$
$\Rightarrow$ (x-axis)

Since tangent at origin is real and coincident (same) the origin is a Cusp.

(iii) Point of intersection

Since curve is symmetric to line $y = 0$
∴ put $y = 0$ in eqn (1) we get $x^3 = 0$
$\Rightarrow x = 0$
Pt. of intersection is (0,0) only

(iv) Asymptotes:

Since $y = \sqrt{\frac{x^3}{(4-x)}}$ tends to ∞ when $x = 4$

∴ line $x = 4$ is an asymptotes to the curve

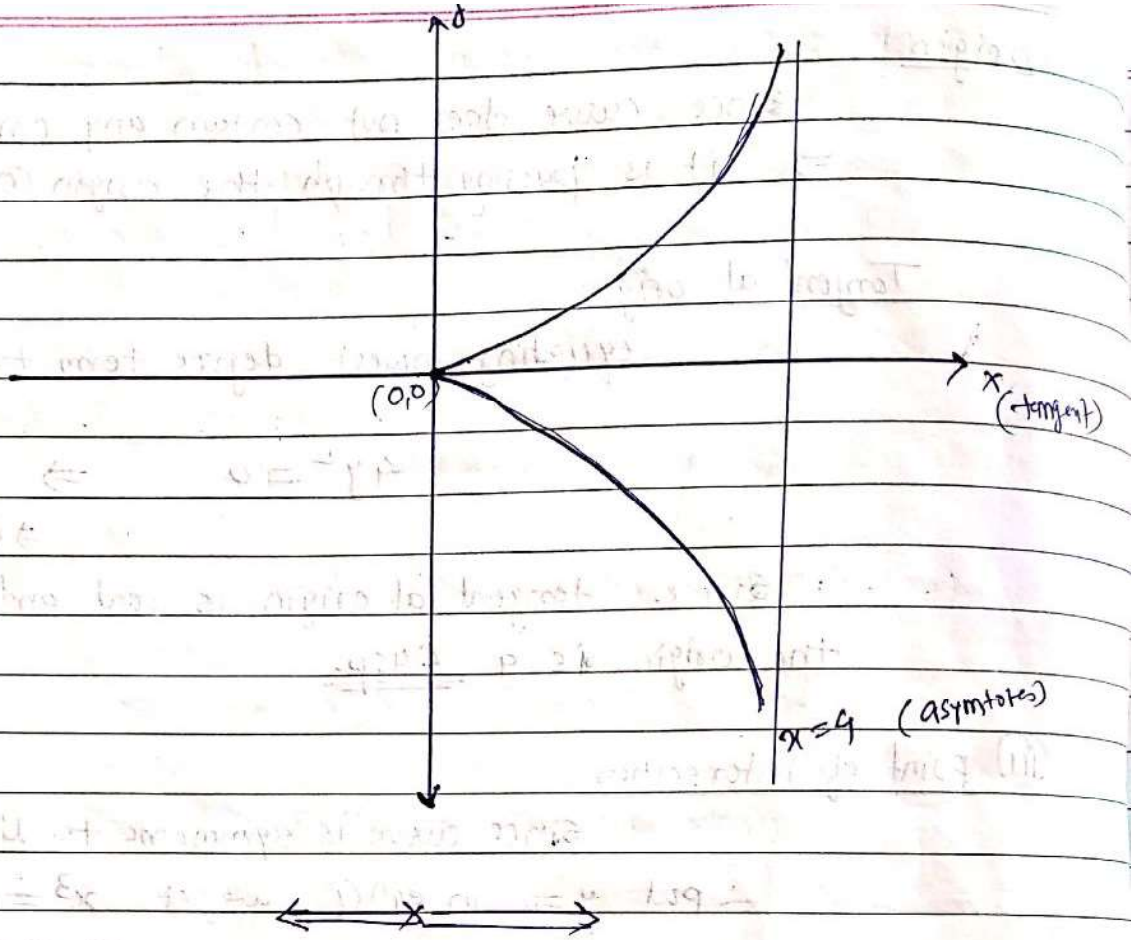
(v) Region: Since $y = \sqrt{\frac{x^3}{(4-x)}}$ become imaginary

if we put $x > 4$ and $x < 0$

∴ Region of existence of a given curve is

$0 < x < 4$

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Ex: Trace the curve

$$xy^2 + (a+x)^2(2a+x) = 0$$

Soln: Given curve is

$$xy^2 + (a+x)^2(2a+x) = 0$$

$$\Rightarrow \quad y = \pm\sqrt{-\frac{(a+x)^2(x+2a)}{x}} \quad \longrightarrow (1)$$

(i) **Symmetry:**

Since given curve contain only even power of y

$\Rightarrow$ it is symmetric to the $y=0$ (i.e. x-axis)

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(ii) Origin : given curve is

$$① \Rightarrow x y^2 + (a^2 + 2ax + x^2)(2a + x) = 0$$

$$\Rightarrow xy^2 + xa^2 + 2ax^2 + x^3 + 2ax^2 + 4a^2x + 2a^3 = 0$$

Since it contain constant term.

$\Rightarrow$ curve is not passing through the origin.

(iii) Point of intersection

Since curve is symmetric to line $y=0$ (x-axis)

$\therefore$ Put $y=0$ in eqn ①, we get

$$① \Rightarrow (a+x)^2 (2a+x) = 0$$

$$\Rightarrow x = -a, \quad x = -2a$$

$\therefore$ point of intersection are $(-a, 0)$ & $(-2a, 0)$.

Tangent at $(-a, 0)$:

put $x = X - a$ and $y = Y + 0$ in eqn ①

$$\Rightarrow (X-a) Y^2 + (X - a + a)^2 (X - a + 2a) = 0$$

$$\Rightarrow XY^2 - aY^2 + X^3 + aX^2 = 0$$

equating lowest degree term to zero

$$a(X^2 - Y^2) = 0$$

$$\Rightarrow Y^2 = X^2$$

$$\Rightarrow Y = \pm X$$

$$\Rightarrow Y = \pm (x + a)$$

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ie. tangents at pt $(-a,0)$ are $y = x+a$ & $y = -(x+a)$

both are real and distinct.

$\therefore$ pt $(-a,0)$ is node.

Tangent at $(-2a,0)$

put $x = X-2a$, $y = Y$ we get

$① \Rightarrow (X-2a)Y^2 + (X-2a+a)^2(X-2a+2a) = 0$

$\Rightarrow XY^2 - 2aY^2 + X^3 - 2aX^2 + a^2X = 0$

equating lowest degree term to zero

$a^2X = 0 \Rightarrow X = 0$

ie. $x+2a = 0$

$\Rightarrow \boxed{x = -2a}$

Since at ~~$x =$~~ $(2a,a)$ we have real and coincident tangent $x = -2a$, it is node.

(iii) Asymptotes:

Since $y = \sqrt{\dfrac{-(x+a)^2(x+2a)}{x}}$ tend to ∞

~~for all values of~~ for $x = 0$.

$\therefore$ line $x = 0$ (y-axis) is an asymptotes to the curve

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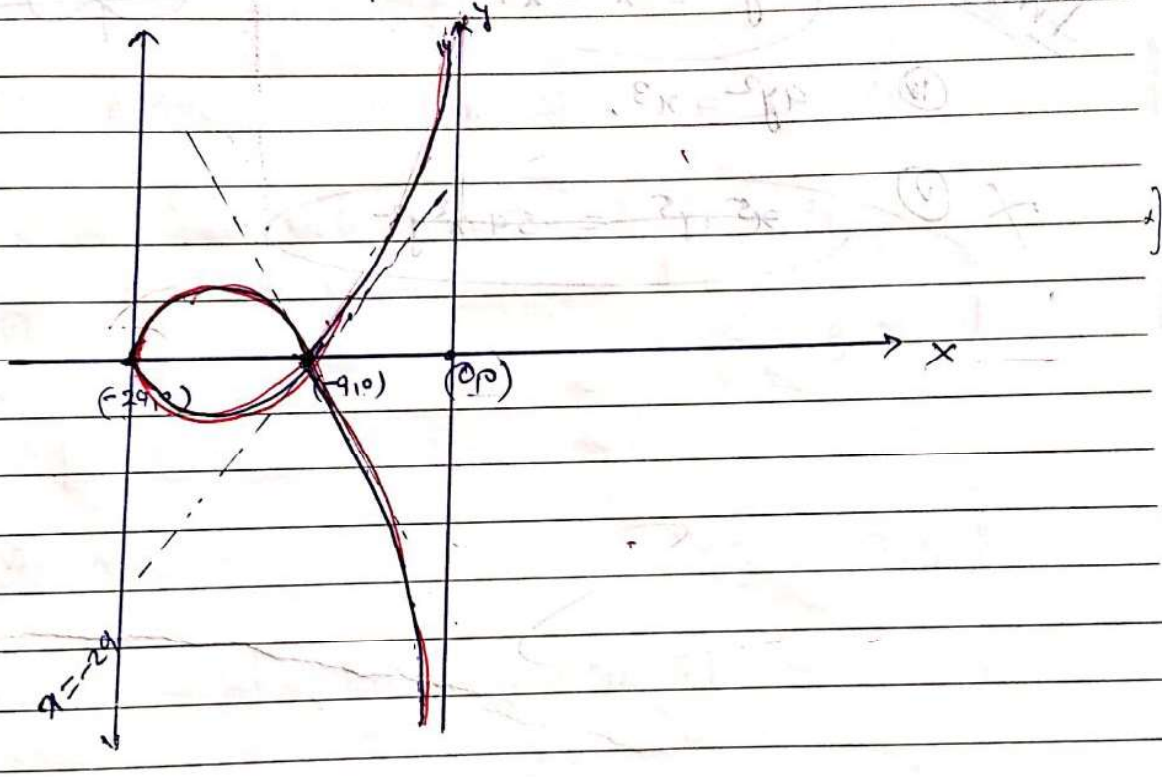
(v) Region. Since

$$y = \sqrt{\frac{-(x+9)^2(x+24)}{x}}$$

become imaginary for all positive values of x

Thus curve is not exist in 1st and 4th quadrants.

i.e. Curve is lies in betn tangent $x = -24$ and asymtotes y-axis. only.



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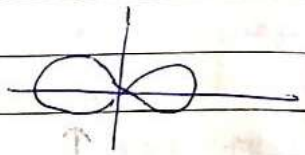
Hw.

Trace à following curve

(i) $a^2x^2 = y^3(2a-y)$

(ii) $ay^2 = x^2(a-x)$

Final (iii) $y^2 = x^2 - x^4$



(iv) $ay^2 = x^3$.

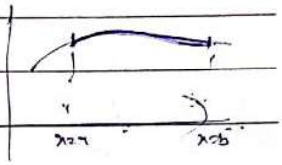
(v) ~~$x^5 + y^5 = 5ax^2y^2$~~

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★ Rectification of a curve (length of a curve).

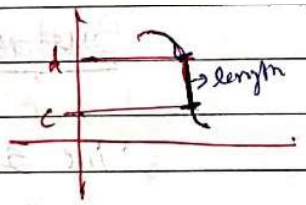
① If $y=f(x)$ be a given curve. then the length of a curve lies betⁿ $x=a$ to $x=b$ is given by

$$S = \int_{x=a}^{b} \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$



② If $x=f(y)$ be the given curve, the length of a curve lies betⁿ $y=c$ to $y=d$ is given by

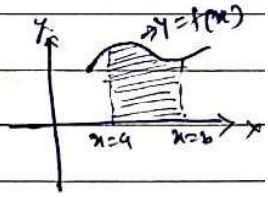
$$S = \int_{y=c}^{d} \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy.$$



(ii) Quadrature (Process of finding ᵃⁿ area bounded by curve).

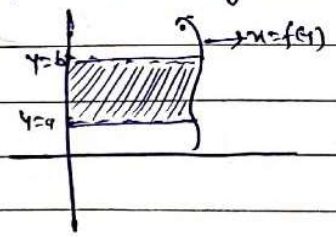
① If $y=f(x)$ be a given curve. then the area bound by curve with line $x=a$ to $x=b$ and x-axis is given by.

$$A = \int_{x=a}^{b} y(x)\, dx$$



② If $x=f(y)$ be a given curve, area bounded by the curve with line $y=c$ to $y=d$ and y-axis is given by

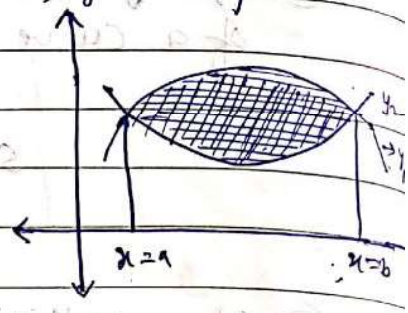
$$A = \int_{y=c}^{y=d} x(y)\, dy$$



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(iii) The area enclosed by two curve $y_1 = f(x)$, & $y_2 = g(x)$ betn the line $x=a$, and $x=b$ given by

$$A = \int_{x=a}^{b} (y_1 - y_2)\, dx$$



Surface of solids of Revolution

The surface area of solid generated by revolving an area of a given curve bounded by $y=f(x)$ by line $x=a$ and $x=b$ is given by

$$\text{Surface Area} = 2\pi \int_{x=a}^{x=b} y \cdot \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$

OR If curve is $x = f(y)$,

$$\text{Surface Area} = 2\pi \int_{y=c}^{y=d} x \cdot \sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy$$

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Volume :

(i) Volume of solid generated by revolution of area bounded by curve $y = f(x)$ with line $x = a$, to $x = b$ about x-axis is given by

$$V = \pi \int_{x=a}^{x=b} y^2 \, dx$$

(ii) Volume of solid generated by revolution of the area bounded the curve $y = f(x)$ with $x = a$ and $x = b$ about x-axis. Line parallel to x-axis ie $y = k$ (say) ~~then~~ given by

$$V = \pi \int_{x=a}^{x=b} (y-k)^2 \, dx$$

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Ex: Find the perimeter of the following curve

$$x^2(a^2-x^2) = 8a^2y^2$$

Soln: Given curve is

$$8a^2y^2 = x^2a^2 - x^4 \quad \text{---(1)}$$

$$\Rightarrow y = \sqrt{\frac{x^2(a^2-x^2)}{8a^2}}$$

Symmetry:

Symmetric to both the co-ordinate axis.

origin:

passing through origin (0,0)

Tangent at origin. $y = \pm x$ ∴ origin is a node

Point of intersection

put $y=0$, (1) $\Rightarrow x=0$ & $x=\pm a$

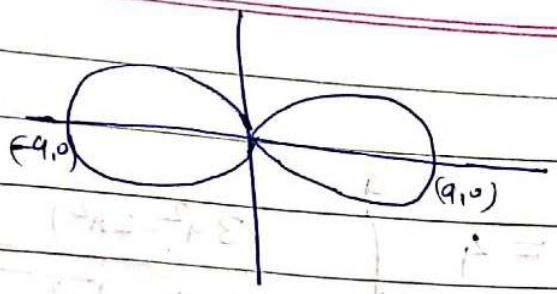
∴ point of intersections are (0,0), (a,0), (-a,0). etc

asymtotes: No. asymtotes

Region: if $x>a$ & $-a<x$ } y become imginam

∴ curve is lies betn $-a < x < a$

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To find perimeter of a given curve:

Diff eqn (1) ~~w.r.t~~ w.r.to x, we get

$$8a^2\left(2y\frac{dy}{dx}\right) = 2a^2x - 4x^3$$

$$\Rightarrow \frac{dy}{dx} = \frac{x(a^2-2x^2)}{8a^2y} = \frac{\sqrt{(8a^2)}\; x(a^2-2x^2)}{8a^2\cdot\sqrt{x^2(a^2-x^2)}}$$

$$\therefore \left(\frac{dy}{dx}\right)^2 = \frac{1}{8a^2}\,\frac{x^2(a^2-2x^2)^2}{x^2(a^2-x^2)} = \frac{(a^2-2x^2)^2}{8a^2(a^2-x^2)}$$

~~length of a~~

perimeter of a curve = total length of curve

$$= 4\left[\text{length of a curve from } x=0 \text{ to } x=a\right]$$

$$= 4\int_0^a \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

$$= 4\int_0^a \sqrt{1+\frac{(a^2-2x^2)^2}{8a^2(a^2-x^2)}}\,dx$$

$$= 4\int_0^a \left(\sqrt{\frac{(3a^2-2x^2)^2}{8a^2(a^2-x^2)}}\right)dx$$

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$$= 4 \int_0^a \frac{(3a^2 - 2x^2)}{2\sqrt{2} \cdot a \sqrt{a^2 - x^2}} dx$$

$$\text{Perimeter} = \frac{\sqrt{2}}{a} \int_0^a \frac{3a^2 - 2x^2}{2\sqrt{2}\sqrt{a^2 - x^2}} dx$$

Put $x = a \sin\theta$.

$\Rightarrow dx = a\cos\theta \, d\theta$

as $x = 0$, $\theta = 0$

$x = a$, $\theta = \pi/2$

$$\text{Perimeter} = \frac{\sqrt{2}}{a} \int_0^{\pi/2} (3a^2 - 2a^2 \sin^2\theta)(a^{-1}\cos^{-1}\theta) \cdot a\cos\theta \, d\theta$$

$$= \sqrt{2}a \int_0^{\pi/2} (3 - 2\sin^2\theta) \, d\theta$$

$$= \sqrt{2}a \int_0^{\pi/2} \left(3 + (\cos 2\theta - 1)\right) d\theta$$

$$= \sqrt{2}a \int_0^{\pi/2} (2 + \cos 2\theta) \, d\theta$$

$$= \sqrt{2}a \left[2\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2}$$

$$\boxed{\text{Perimeter} = \sqrt{2}a\pi}$$

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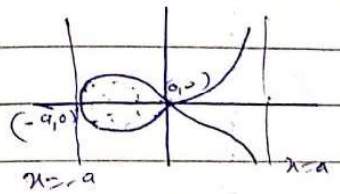
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Ex: Find the volume of a solid of revolution obtained by revolving the loop of the curve

$$y^2 = x^2\left[\frac{a+x}{a-x}\right]$$ about x-axis

Hint:

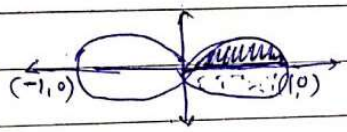


$$\text{Volume} = \pi \int_{-a}^{0} y^2\,dx = \pi \int_{-a}^{a} \frac{x^2(a+x)}{(a-x)}\,dx$$

$$= 2\pi a^3\left[\log 2 - \tfrac{2}{3}\right]$$ put $(a-x) = t$

Ex: Trace the curve $y^2 = x^2 - x^4$ and find the area of the one of it's loop.

Hint:

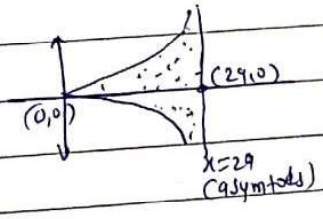


Area of one of the loop. $= 2\int_{x=0}^{1} y\,dx$

$= 2/3$

Ex Trace the curve $y^2(2a-x) = x^3$, Also find the area enclosed betn the curves and it's asymptotes

Hint



$$\text{Reqd area} = 2\int_{x=0}^{2a} y\,dx = 2\int_{0}^{2a}\left(\frac{x^3}{2a-x}\right)^{1/2}$$

$$= 2\int_{0}^{2a} \frac{x^{3/2}}{(2a)^{1/2}\left(1-x/2a\right)^{1/2}}\,dx$$

put $x/2a = t$

$$\therefore \text{Area} = 8a^2\int_{0}^{1} t^{3/2}(1-t)^{-1/2}\,dt$$

$$= 8a^2\,\beta\left(\tfrac{5}{2}, \tfrac{1}{2}\right)$$

$A = 3a^2\pi$ sq.unit.

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